

MASTER

MODELLING, DATA ANALYSIS AND DECISION SUPPORT SYSTEMS

Sales Forecasting in Fast Fashion Industry

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Dissertation

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Abstract

Replenishment is a vital operation in supply chain and inventory management, acting as a crucial link between accurate demand forecasting and effective stock allocation. In the dynamic fashion retail industry, this process is particularly intricate due to the sector's distinctive characteristics, such as short product lifecycles, high product turnover and rapidly changing consumer preferences. Moreover, factors like seasonality and unpredictable trends intensify these challenges, forcing retailers to adapt quickly to fluctuations in demand while minimising inefficiencies. Maintaining optimal stock levels is essential not only to prevent overstocking, which can lead to increased holding costs and markdowns, but also to reduce stockouts, which may result in lost sales opportunities and decreased customer satisfaction. When supported by precise forecasting techniques, replenishment enables businesses to navigate these complexities, streamline operations and make agile, data-driven decisions that align with market dynamics.

This dissertation aims to address these challenges by developing an advanced replenishment model. In collaboration with a Portuguese fashion retail company, this project intends to optimise stock allocation and improve inventory management practices by utilising advanced forecasting techniques.

A variety of forecasting methods, encompassing traditional statistical models, count-based techniques, machine learning algorithms and hybrid approaches were evaluated in this project's empirical analysis. Among these, boosting techniques consistently delivered the highest accuracy, with CatBoost emerging as the top-performing model by achieving the lowest values for MAE, RMSE and MAAPE. When compared to the company's existing replenishment method, the boosting models displayed a significant reduction in forecasting errors and offered a more balanced error profile, characterised by a higher number of exact matches and fewer extreme deviations. These findings underscore the potential of advanced machine learning techniques to enhance stock allocation, mitigate inefficiencies and facilitate more agile, data-driven decision-making within the fast fashion retail sector.

Keywords: Sales forecasting; replenishment process; fashion retail; time series; machine learning.

Resumo

A reposição de produtos é uma operação vital na gestão da cadeia de abastecimento e de inventário, funcionando como ligação crucial entre a previsão precisa da procura e a alocação eficaz de stock. No dinâmico setor do retalho de moda, este processo é complexo devido a ciclos de vida curtos, elevada rotatividade de stock e rápidas mudanças nas preferências dos consumidores. Além disso, fatores como a sazonalidade e as tendências imprevisíveis intensificam estes desafios, obrigando os retalhistas a adaptarem-se rapidamente às flutuações da procura, minimizando ineficiências. Manter níveis ótimos de stock é essencial para evitar excessos, que podem levar a custos acrescidos de armazenagem e reduções de preço, mas também para reduzir ruturas de stock, que resultam em perdas de vendas e diminuição da satisfação do cliente. Quando apoiada por técnicas de previsão rigorosas, a reposição permite que as empresas enfrentem estas complexidades, otimizem operações e tomem decisões orientadas por dados, em linha com a dinâmica do mercado.

Esta dissertação procura responder a estes desafios através do desenvolvimento de um modelo avançado de reposição. Em colaboração com uma empresa portuguesa de retalho de moda, o projeto pretende otimizar a alocação de stock e melhorar as práticas de gestão de inventário recorrendo a técnicas avançadas de previsão.

Uma variedade de métodos de previsão, que abrangem modelos estatísticos tradicionais, técnicas baseadas em contagem, algoritmos de *machine learning* e abordagens híbridas foram avaliados na análise empírica deste projeto. Entre estes, as técnicas de *boosting* demonstraram consistentemente a maior precisão, destacando-se o *CatBoost* como o modelo com melhor desempenho, ao alcançar os valores mais baixos de MAE, RMSE e MAAPE. Quando comparados com o método de reposição atualmente utilizado pela empresa, os modelos de *boosting* revelaram uma redução significativa nos erros de previsão e um perfil de erro mais equilibrado, caracterizado por um maior número de previsões exatas e menos desvios extremos. Estes resultados evidenciam o potencial das técnicas avançadas de *machine learning* para melhorar a alocação de stock, mitigar ineficiências e promover decisões mais ágeis e orientadas por dados no setor do retalho de moda rápida.

Palavras-chave: Previsão de vendas; processo de reposição; retalho de moda; séries temporais; *machine learning*.

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Glossary

ADF - Augmented Dickey-Fuller test

AI - Artificial Intelligence

ANN – Artificial Neural Networks

ARMA - Autoregressive Moving Average

ARIMA - Autoregressive Integrated Moving Average

ELM – Extreme Learning Machine

FRSC – Fashion Retail Supply Chain

FRSCM – Fashion Retail Supply Chain Management

GBM – Gradient Boosting Machines

GLM - Generalised Linear Model

INGARCH - Integer-Valued Generalised Autoregressive Conditional Heteroskedasticity

JIT - Just-In-Time

KPSS - Kwiatkowski–Phillips–Schmidt–Shin test

NB – Negative Binomial

MAAPE - Mean Arctangent Absolute Percentage Error

MAE - Mean Absolute Error

MAPE - Mean Absolute Percentage Error

PTT - Probability Integral Transform

QR - Quick Response

RMSE - Root Mean Squared Error

SARIMA - Seasonal Autoregressive Integrated Moving Average

SMAPE - Symmetric Mean Absolute Percentage Error

SMAPE3 - Symmetric Mean Absolute Percentage Error (refined version)

SKU - Stock Keeping Units

SVM – Support Vector Machines

WAPE - Weighted Absolute Percentage Error

ZINB - Zero-Inflated Negative Binomial

ZIP - Zero-Inflated Poisson

1. Introduction

The fashion industry is characterised by intense competition, driven by globalisation, rapidly changing consumer preferences and the need for operational efficiency. In this context, sales forecasting is crucial for retail and supply chain management, enabling companies to optimise inventory and respond effectively to market demands. However, the forecasting process in this sector is highly complex due to several factors, including short product lifecycles, a wide variety of products and unpredictable consumer behaviour (Şen, 2008). Additionally, challenges like seasonality, volatile market conditions and the constant introduction of new collections lacking historical data make accurate predictions even more difficult (Thomassey, 2016).

Despite these challenges, robust forecasting systems can significantly enhance the overall performance of companies in the fashion sector. They help to prevent both understocking and overstocking, ensuring efficient inventory management and supporting critical supply chain operations, such as production planning, pricing strategies and customer satisfaction (Frank et al., 2003; Liu et al., 2013). Thus, accurate sales forecasting enables companies to navigate the market complexities, reduce costs and enhance their competitiveness.

1.1. Context

The retail sector, especially within the fashion industry, has witnessed a significant growth in innovation and competitiveness. This trend is fuelled by changing market demands, globalisation and the rapid adoption of advanced technologies (Christopher, 2022). In this dynamic environment, companies encounter a crucial challenge: effectively managing stock allocation to ensure the availability of the right products at the right time and in the right locations. This challenge is particularly acute in the fast fashion segment, where short product lifecycles and unpredictable consumer behaviour intensify the complexity of inventory management (Wen et al., 2019).

Effective forecasting and optimised replenishment strategies can provide substantial advantages to companies. By leveraging advanced predictive models, organisations can significantly reduce the costs associated with overstocking and stockouts, improve supply

chain efficiency and enhance customer satisfaction by ensuring the availability of key products (Caro & Gallien, 2010; Şen, 2008). These improvements can lead to more efficient stock distribution across stores, better alignment with market trends and higher inventory turnover, ultimately increasing profitability and customer loyalty (Cachon & Swinney, 2011). The primary aim of this study is to deliver effective solutions that address these challenges while aligning with the company's strategic objectives.

While forecasting offers a wealth of potential benefits, it also poses several challenges. Accurately predicting demand in the fashion industry is particularly complex due to factors such as erratic consumer behaviour, seasonal fluctuations and the introduction of new collections that lack historical data (Liu et al., 2013; Thomassey, 2016). Furthermore, external factors, namely economic shifts and unforeseen trends, can have a significant impact on the accuracy of forecasts. These challenges highlight the necessity of robust and adaptable models that can provide reliable insights to support effective decision-making, enabling the company to navigate the complexities of the market while maintaining a competitive edge.

1.2. Motivation and Objectives

This project aims to address a significant challenge faced by a Portuguese fashion retail company by answering the following research question: “How can advanced sales forecasting methods be used to enhance replenishment decisions in fashion retail, aiming to reduce stockouts, minimise overstocks and improve inventory efficiency?”.

The primary objective of this study is to offer actionable insights and practical solutions to optimise the replenishment process, enabling the company to allocate products effectively across its global store network. This will involve evaluating and comparing the performance of various sales forecasting methods, simulating the implementation of the most effective model in the replenishment process and providing clear and concise recommendations tailored to the company's specific context and operational objectives.

The company currently uses a replenishment forecasting method based on historical data. This approach analyses sales records from the past 7 or 14 days, depending on the country, to estimate future demand. The sales velocity, calculated as the quantity sold divided by the number of days (e.g., weekly sales / 7), serves as the basis for predicting the sales rate during the forecasting period. To enhance these predictions, a multiplier is applied to account

for seasonal variations. While this method provides a straightforward and operational approach to generating short-term forecasts, it is inherently reactive and heavily dependent on very recent trends. Consequently, it may overlook deeper, non-linear patterns and fail to anticipate shifts in consumer behaviour prompted by external factors, such as marketing campaigns, holidays or fluctuations in product availability. These limitations underscore the importance of exploring more advanced forecasting methodologies, particularly those capable of integrating additional variables, learning from historical patterns and offering enhanced adaptability to the rapidly changing landscape of fashion retail.

1.3. Dissertation Structure

The rest of this dissertation is organised into four chapters, each addressing a key phase of the research.

The second chapter, entitled “Literature Review”, offers a thorough examination of the key concepts underpinning this study, particularly within the context of the fashion retail industry and its associated forecasting methods. It begins by analysing the unique characteristics of the sector, including the structure of its supply chain, the nuances of fast fashion operations and the effects of seasonality and rapidly changing trends. Furthermore, it addresses the primary challenges related to inventory management in this environment, emphasising the necessity of balancing stock availability with minimal excess. The chapter also introduces the foundational principles that govern replenishment practices in fashion retail.

The latter part of the chapter delves into forecasting methods, beginning with an overview of traditional time series techniques used for sales predictions. It then investigates more advanced approaches, like machine learning models and hybrid strategies that merge statistical rigour with data-driven adaptability. Special focus is given to forecasting methods tailored for intermittent demand and count time series, which are particularly pertinent in the fashion retail sector due to the prevalence of products with irregular or low sales frequency. The chapter also discusses evaluation metrics that are especially relevant for replenishment-oriented forecasting, encompassing both statistical accuracy measures and business-oriented indicators, and concludes with a review of cross-validation approaches specifically designed for time series data.

The third chapter, titled “A CRISP-DM Based Framework for Sales Forecasting”,

outlines the practical development of the project in alignment with the CRISP-DM methodology, which offers a structured and iterative approach to building data mining solutions. The chapter opens with the Business Understanding phase, where the forecasting needs of the fashion retail company are detailed, alongside the primary challenges related to demand variability and intermittent sales. This is followed by the Data Understanding phase, which investigates the structure, granularity and initial characteristics of the sales and product datasets provided by the company. The Data Preparation section describes the data cleaning procedures implemented and the initial transformations applied to the datasets.

The Modelling phase is organised into several essential components. It begins with the data-splitting strategy and the establishment of the forecasting approach, which ensures methodological rigour in managing time-dependent information. Subsequently, traditional statistical models, machine learning algorithms and hybrid forecasting strategies are presented, each contributing uniquely to the management of intermittent and low-volume demand. The chapter then delves into the feature engineering process, where temporal, product-level and store-level attributes are integrated into the dataset, alongside the tuning procedures applied to the boosting models. Lastly, the chapter outlines a cross-validation method specifically designed for time series, offering a robust framework for evaluating both accuracy and stability.

The fourth chapter, titled "Results and Evaluation", presents a comparative analysis of the forecasting models using established evaluation metrics and error histograms. It offers statistical accuracy measures and insights into the distribution of errors. This chapter merges quantitative assessment with critical interpretation, emphasising the relative strengths and weaknesses of each approach. Furthermore, the best-performing models are compared to the company's current forecasting method, enabling an evaluation of their practical relevance in enhancing demand forecasting and informing replenishment decisions.

The fifth chapter, entitled "Conclusions", offers a thorough summary of the research's key findings, emphasising the objectives met during the course of the project. It is organised into three sections: the first delineates the main achievements of the study, the second addresses the limitations encountered, including both methodological and data-related challenges, and the third presents recommendations for future research. This section suggests avenues for expanding the forecasting model to encompass additional product categories and store profiles, as well as investigating further variables and techniques to enhance accuracy and optimise inventory within the fashion retail sector.

2. Literature Review

This chapter aims to investigate the essential concepts and methodologies that underlie sales forecasting and inventory management in the fashion retail sector. It begins by analysing the supply chain dynamics prevalent in this fast-paced industry, highlighting the unique challenges posed by the unpredictable nature of consumer demand and the short lifecycles of fashion products. The discussion then moves on to the principles of fast fashion, emphasising the significance of speed, flexibility and innovation in shaping the operational strategies of retail companies. Additionally, the influence of seasonality and trends is explored, showcasing their impact on consumer behaviour and the complexities they introduce into demand forecasting.

The chapter also addresses the complexities of inventory management, including the trade-offs between overstock and stockouts and examines the key principles of replenishment that guide stock distribution. The chapter ultimately delves into forecasting practices within the fashion industry, presenting a comparative analysis of traditional time series models and contemporary machine learning techniques, with particular attention to their application in daily sales forecasting and their adaptability to count-based data.

It places particular emphasis on the nuances of daily sales forecasting and the application of models specifically designed for count time series, which are prevalent in this sector due to the occurrence of frequent zero-sales periods. The chapter also reviews evaluation metrics tailored for replenishment forecasts, focusing on both statistical accuracy and business relevance. Finally, it introduces cross-validation approaches adapted to time series, which are essential for assessing model robustness. Together these discussions provide a comprehensive foundation for the methodological choices made in the subsequent chapters.

2.1. Fashion Retail

The fashion retail industry is characterised by significant volatility, short product lifecycles and ongoing pressure to align supply with swiftly changing consumer preferences. These traits present unique operational challenges that require flexible, responsive and data-driven strategies. This section offers an overview of the structural and strategic foundations of the sector, beginning with the organisation of its supply chain and advancing through the

fast fashion model, the impact of seasonality and trends, and the fundamental challenges of inventory management. The discussion concludes by examining replenishment principles, which are essential for ensuring stock availability while minimising inefficiencies in this competitive and dynamic retail landscape.

2.1.1. Supply Chain

In today's business landscape, supply chain management plays a pivotal role in ensuring operational efficiency and driving overall success. A supply chain encompasses the entire network of processes and entities involved in transforming raw materials into finished products and delivering them to consumers (Mentzer et al., 2001). This system requires seamless coordination among suppliers, manufacturers, distributors and retailers to maintain a steady flow of goods, information and financial resources (Christopher, 2022).

An agile and well-structured supply chain provides companies with a critical competitive advantage by enabling them to adapt quickly to changing customer demands and market fluctuations. This adaptability is particularly crucial in industries with rapidly changing consumer preferences, where reducing lead times and optimising inventory levels are essential (Camargo et al., 2020). Efficient supply chains ensure that the right products reach the right place at the right time, supporting high service levels and minimising operational inefficiencies.

Within this broader framework, the fashion retail supply chain (FRSC) stands out as a dynamic and consumer-driven system, characterised by the rapid flow of products, information and funds. The short product lifecycles and ever-changing consumer trends place significant demands on FRSCs, making them inherently complex to manage (Agrawal & Smith, 2015; Choi, 2014). Fashion retail supply chain management (FRSCM) integrates sourcing, production and logistics to align supply with volatile demand and ensure that customer needs are met efficiently (Council of Supply Chain Management Professionals [CSCMP], n.d.; Choi & Cheng, 2015).

The importance of FRSCM lies in its ability to balance speed, quality and cost-effectiveness. It relies on accurate demand forecasting, real-time information flow and competitive pricing strategies to maintain operational efficiency and meet customer expectations in a highly competitive market (Hanne et al., 2017; Şen, 2008).

Key features of the FRSC include short lead times and operational flexibility. Fast fashion strategies that focus on reducing lead times allow retailers to respond swiftly to

changing market demands (Cachon & Swinney, 2011; Choi & Sethi, 2010). Additionally, flexibility enables supply chains to adjust to seasonal fluctuations and unpredictable trends, ensuring timely and efficient product delivery (Wen et al., 2019).

However, managing an FRSC comes with its challenges. Effective coordination among suppliers, distribution centres and retail stores is critical to synchronise inventory levels and respond promptly to consumer demand (Choi & Cheng, 2015). Furthermore, global disruptions, such as natural disasters, political instability and pandemics, can severely impact supply chain operations, resulting in delays, increased costs and unmet demand (Wen et al., 2019).

Despite these challenges, FRSCM remains fundamental to the success of fashion retailers, enabling them to navigate the complexities of the industry and maintain their competitive edge through efficient and responsive operations.

2.1.2. Fast Fashion Retail

Fast fashion is generally characterised as a business model that emphasises quick product development and frequent market introductions, allowing retailers to deliver trendy items to consumers in the shortest possible timeframe. This strategy focuses on continuously refreshing store assortments with small production runs, ensuring a high level of product variety and sustaining consumer engagement (Mehrijoo & Pasek, 2016; Thomassey, 2016).

A pivotal factor contributing to the success of fast fashion is the Quick Response (QR) strategy, which effectively reduces lead times and improves demand forecasting. By implementing QR systems, supply chains can swiftly adapt to market fluctuations, thereby minimising the risks of stockouts and overproduction (Wen et al., 2019). For example, Zara, a prominent player in the fast fashion industry, has streamlined its supply chain to complete the entire production cycle, from design to retail shelves, in just two weeks. This contrasts dramatically with the traditional apparel industry, where the typical timeframe is six to nine months (Ghemawat, 2003).

This business model poses significant challenges, particularly in the areas of demand forecasting and inventory management. The high volatility of demand, influenced by ever-changing trends and consumer preferences, hinders the accurate prediction of sales. Additionally, the short product lifecycles characteristic of fast fashion restrict the availability of historical data for individual Stock Keeping Units (SKUs), further exacerbating the difficulty of forecasting demand and increasing the risk of both overstocking and stockouts

(Choi et al., 2013).

The emergence of digital platforms has intensified these challenges by facilitating impulsive purchasing behaviour among consumers. E-commerce and social media significantly influence the fast fashion industry, as they expedite trend dissemination and generate unpredictable spikes in demand. This digital impact compels fast fashion retailers to improve their agility and leverage advanced analytics to better forecast consumer behaviour and optimise inventory management.

From an operational perspective, fast fashion also places significant pressure on manufacturers, who must contend with reduced order quantities and high costs associated with building flexible supply chains (Mehrjoo & Pasek, 2016). To address these issues, strategic contracts, including inventory service level agreements, buy-back contracts and minimum order quantity commitments, are often implemented to ensure smooth operations and risk-sharing across the supply chain (Choi & Chow, 2008).

Despite the inherent complexities, fast fashion continues to prosper by capitalising on its speed, flexibility and its ability to meet the modern consumer's desire for instant gratification and constant novelty. Nevertheless, the long-term sustainability of this model will hinge on its ability to harmonise operational efficiency with the increasing demand for more ethical and environmentally friendly practices.

2.1.3. Seasonality and Trends

The fashion industry faces significant challenges in forecasting demand, primarily due to the unpredictability of trend peaks and seasonal fluctuations.

Seasonality plays a crucial role in shaping demand patterns, with specific periods such as Spring/Summer and Fall/Winter influencing consumer behaviour. In order to make accurate predictions, sales forecasting systems must incorporate these seasonal elements (Thomassey, 2016). For example, items like swimwear and pullovers are particularly sensitive to seasonal changes, whereas basic undergarments tend to maintain consistent demand throughout the year. This variability highlights the necessity for forecasting systems that can adapt to the unique seasonal sensitivities of different product categories, thereby ensuring effective inventory management and replenishment strategies.

In addition to seasonality, fashion trends have a significant impact on consumer preferences, often resulting in short and volatile demand cycles. These trends, driven by rapid shifts in consumer tastes and the impact of social media, add another layer of complexity to

demand forecasting. Unlike seasonality, which generally follows more predictable patterns, trends are more erratic and difficult to anticipate (Choi et al., 2013). This underscores the importance of agile forecasting methods that can respond swiftly to emerging trends, minimising the risk of overstock and stockouts.

2.1.4. Inventory Management Challenges in Fashion Retail

Inventory management in the fashion retail sector poses unique challenges due to the unpredictable nature of consumer demand and the short lifecycles of products. One of the most pressing issues is the balance between overstock and stockouts. Overstock can lead to increased holding costs, diminished profitability and the necessity for markdowns to clear unsold merchandise. Conversely, stockouts result in lost sales opportunities and customer dissatisfaction, which can ultimately undermine brand loyalty (Caro & Gallien, 2010). Achieving this balance requires accurate sales forecasting, since effective inventory planning depends on anticipating future demand to align supply with customer needs (Liu et al., 2013).

The fast-paced environment of the industry also contributes to high product turnover, requiring frequent restocking to keep up with rapidly evolving trends. Short product lifecycles demand responsive supply chains and real-time inventory tracking to ensure that stock levels match demand without incurring excess. Inefficiencies in this process, such as delays in replenishment or inaccurate demand forecasts, can lead to financial losses and missed opportunities (Tokatli, 2008).

Additionally, inventory strategies must be customised to the specific characteristics of seasonal and permanent products. Seasonal items, like winter coats or summer swimwear, are highly sensitive to trends and require precise forecasting to minimise unsold inventory at the end of the season. In contrast, permanent products, such as basic T-shirts or undergarments, tend to have more consistent demand over time, allowing for more stable inventory levels (Thomassey, 2016). Successfully managing these distinct categories requires a flexible and data-driven approach to inventory planning.

2.1.5. Replenishment Principles

Replenishment is one of the most critical operations within the FRSC, playing a vital role in maintaining optimal inventory levels and effectively addressing customer demand. As noted by Thomassey (2010), the supply strategy in this sector generally encompasses two

main phases: the initial long-term distribution of products to stores at the beginning of the sales season, followed by short-term replenishment throughout the season. This two-phase approach enables retailers to respond dynamically to sales performance, ensuring that popular items are restocked while simultaneously minimising excess inventory.

Short-term replenishment is particularly crucial in fast-moving markets such as fashion, where trends evolve rapidly and product lifecycles are short. Accurate forecasting during this phase is essential to align inventory with real-time demand, thereby avoiding costly inefficiencies like overstocking and stockouts. Moreover, timely replenishment fosters a more agile supply chain, providing retailers with a competitive advantage in a volatile market (Du et al., 2015).

Different product categories require distinct replenishment strategies. Seasonal items, which are significantly influenced by trends, require precise inventory planning in order to capture peak demand without resulting in surplus stock. In contrast, permanent products, characterised by a more consistent sales pattern, benefit from steady and predictable replenishment cycles. Advanced forecasting tools, including neural networks and optimisation models, have demonstrated their effectiveness in enhancing replenishment accuracy, enabling retailers to optimise their supply chain operations (Du et al., 2015).

2.2. Forecasting Methods in Fashion Retail

Sales forecasting plays a vital role in the strategic and operational decision-making processes of fashion retailers. Due to the industry's rapid product turnover, brief selling windows and the constant evolution of trends, accurately predicting consumer demand is both complex and essential. Reliable forecasts facilitate inventory optimisation, mitigate the risks of overstock and stockouts and improve replenishment strategies. To tackle these challenges, a diverse array of forecasting methods has been developed, ranging from traditional time series models that leverage historical patterns to more sophisticated approaches employing machine learning and hybrid techniques. This section delves into these forecasting methods, focusing on their relevance for daily demand prediction and their effectiveness with intermittent and count-based retail data.

2.2.1. Time Series Forecasting

Time series forecasting constitutes a central approach in predictive analytics, aiming

to predict future values by modelling the temporal patterns embedded in historical data. This approach is widely used across various industries where understanding temporal dynamics is essential for strategic decision-making. By examining time series data, researchers and practitioners aim to identify underlying patterns such as trends, seasonality and randomness. These insights help to develop accurate forecasting models (Box et al., 2015; Hyndman & Athanasopoulos, 2018).

A time series can be divided into four main components: trend, seasonality, cyclic patterns and randomness. According to Box et al. (2015), Holt (1957), Hyndman and Athanasopoulos (2018) and Makridakis et al. (2008), these components are defined as follows:

- **Trend:** the long-term direction of the data, indicating whether it is moving upward, downward or remaining stable over time.
- **Seasonality:** recurring patterns that occur at fixed intervals, such as daily, weekly, or annual fluctuations. Seasonality is particularly significant in industries like fashion retail, where demand often follows seasonal cycles.
- **Cyclic Patterns:** irregular oscillations that occur over longer periods and are typically influenced by external factors, like economic cycles.
- **Randomness:** unpredictable variations in the data that cannot be attributed to a specific pattern.

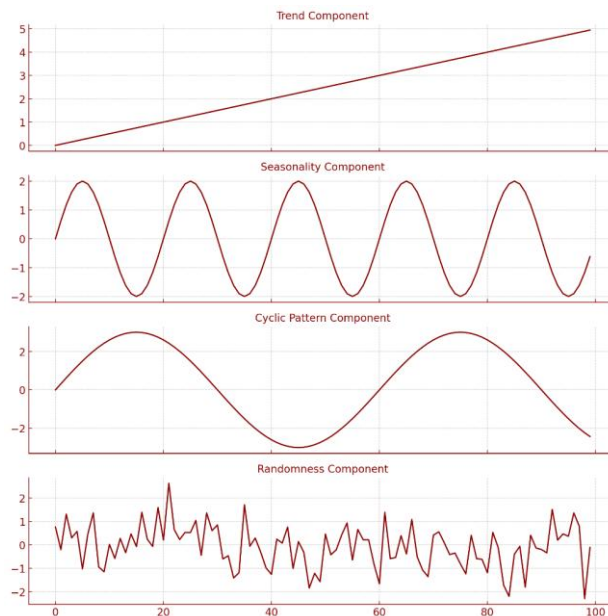


Figure 1: Components of a time series – trend, seasonality, cyclic pattern and randomness. Source: Inspired by Hyndman & Athanasopoulos (2021), *Forecasting: Principles and Practice* (3rd ed.). Retrieved from <https://otexts.com/fpp3/>

Another key concept in time series analysis is stationarity, which denotes the characteristic of a series that maintains a constant mean and variance over time. Additionally, in a stationary series, the serial correlation depends only on the lag, allowing for the meaningful calculation of statistics like mean, variance and quartiles. Many traditional forecasting models, namely Autoregressive Moving Average (ARMA), assume stationarity as a prerequisite for making accurate predictions. However, models such as Autoregressive Integrated Moving Average (ARIMA) include a differencing step to handle non-stationary data and Exponential Smoothing methods do not require stationarity to produce reliable forecasts (Box et al., 2015; Hyndman & Athanasopoulos, 2018). When a series exhibits non-stationarity, due to factors like trends, seasonality or external shocks, transformation techniques such as differencing or logarithmic adjustments are often used to stabilise the data. To evaluate whether a series demonstrates stationarity, statistical tests including the Augmented Dickey-Fuller (ADF) and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) are frequently utilised (Gardner Jr, 2006).

Time series analysis offers a systematic approach to understanding and modelling data that varies over time. By decomposing the data into its essential components and ensuring stationarity, time series methods lay the groundwork for forecasting models that are widely used in various industries. These techniques allow us to use historical patterns to predict future outcomes, making them essential for creating accurate and reliable predictive models.

2.2.2. Advanced Forecasting Practices: A Comparative Analysis

The field of sales forecasting in fashion retail has attracted considerable academic and practical interest, largely due to the unique challenges inherent to this dynamic industry. A variety of studies have investigated innovative methodologies for predicting demand in this sector. This research attempts to address the complexities of forecasting in fashion retail, such as integrating external variables, managing new product launches without extensive historical data and adapting to seasonal trends and unpredictable market conditions.

Thomassey (2016) proposed a hybrid strategy designed to tackle the unique challenges of the fast fashion sector by integrating traditional methodologies with machine learning in a two-step forecasting system. The first step involved developing a long-term model using clustering and decision trees to predict sales for new products, even in the absence of historical data. Clustering was used to group products with similar characteristics, facilitating the transfer of sales patterns from comparable items. Meanwhile, decision trees

were used to identify the key factors influencing sales, including category, material and seasonality. In a second step, a short-term model was introduced using Extreme Learning Machine (ELM), a machine learning technique that refines initial forecasts by incorporating actual sales data from the first two weeks. This dual approach allowed for more precise predictions and quicker responses to changes in demand. The use of traditional methods in the first step provided a stable foundation for initial forecasts, while the integration of machine learning in the second step ensured enhanced accuracy and adaptability in the dynamic and volatile fast fashion landscape.

Martins and Galegale (2022) conducted an integrative literature review to explore the application of various sales forecasting methods in the retail sector, contrasting traditional techniques with machine learning algorithms. Their analysis encompassed a broad spectrum of studies, revealing that the most frequently used traditional methods include ARIMA and Exponential Smoothing, while machine learning techniques feature Artificial Neural Networks (ANN), Support Vector Machines (SVM) and Random Forests. The authors highlighted that although traditional methods are effective in stable markets due to their simplicity and cost-effectiveness, machine learning algorithms exhibit superior accuracy when dealing with exogenous variables and intricate demand patterns. The review emphasises the advantages of machine learning based methods in dynamic environments while acknowledging that the choice of forecasting technique depends on the specific context and available resources.

Hyndman and Athanasopoulos (2018) offer a comprehensive discussion on Exponential Smoothing techniques, with a particular focus on Holt-Winters methods, which are extensively utilised for forecasting time series data that exhibit trends and seasonality. These methods enhance simple exponential smoothing by incorporating components for trend and seasonal adjustments, thus enabling more adaptable modelling in structured retail environments. However, the authors caution that Holt-Winters methods may not perform optimally when applied to highly irregular and zero-inflated datasets often seen in the fashion retail industry, as they tend to be sensitive to abrupt demand fluctuations and non-repetitive patterns.

Anitha and Neelakandan (2024), Swaminathan and Venkitasubramony (2024) and Liu et al. (2013) have conducted extensive reviews of demand forecasting methods, each focusing on different aspects of the fashion retail sector. Despite their unique perspectives, all of these studies emphasise the importance of using a combination of traditional methods,

machine learning techniques and hybrid models to address the complexity and volatility of demand in the fashion industry.

Traditional methods such as regression, ARIMA and Exponential Smoothing are acknowledged for their simplicity and effectiveness in stable environments; however, they often struggle with non-linear patterns and the integration of external variables like weather and trends. In contrast, machine learning techniques, including ANN, SVM and ELM, excel at capturing complex nonlinear relationships and incorporating external factors, making them particularly valuable in dynamic and unpredictable markets. Additionally, these studies highlight the increasing relevance of hybrid models, which combine the robustness of traditional approaches with the adaptability of machine learning algorithms. Hybrid models are characterised by their ability to leverage the strengths of each approach, often using traditional methods for initial forecasts and refining them with machine learning techniques.

Teunter et al. (2011) investigate the effectiveness of Croston's method for managing intermittent demand, a common challenge in the fashion retail sector. The study highlights how traditional forecasting models often struggle to accommodate sparse and irregular demand patterns, leading to less-than-optimal inventory decisions. In contrast, Croston's method effectively addresses this issue by separately estimating the demand size and the intervals between non-zero sales, thereby providing a customised solution for highly intermittent time series. The authors conclude that Croston's method proves valuable in situations characterised by inconsistent product availability and erratic purchasing patterns, making it particularly suitable for the dynamics of fast fashion environments.

Freeland and McCabe (2004) focus on the use of Poisson Autoregressive (PAR) models for count data forecasting. The PAR technique is specifically designed for low-count series, capturing the number of events occurring within a fixed period of time. The authors emphasise the challenges associated with using the conditional mean in low-count series, proposing the conditional median as a more robust alternative. This approach is particularly useful when dealing with intermittent sales data.

Kolassa (2016) addresses the problem of evaluating predictive distributions for count data forecasting, especially in retail environments where sales can vary significantly between periods. He explores several models suited for count data, including Poisson, Negative Binomial (NB) and Zero-Inflated models, such as Zero-Inflated Poisson (ZIP) and Zero-Inflated Negative Binomial (ZINB). Kolassa advocates for the use of scoring rules and predictive distributions, namely the Probability Integral Transform (PIT), to assess model

accuracy. These methods are especially relevant when dealing with high variability and low demand levels.

Liboschik et al. (2017) introduce the R package *tscount*, which supports a range of count time series models, including Poisson Generalised Linear Models (GLM), Negative Binomial GLM, ZIP, ZINB and Integer-Valued Generalised Autoregressive Conditional Heteroskedasticity (INGARCH). The INGARCH model, in particular, captures serial dependence in count data by incorporating past observations and past means. These models are particularly relevant for fashion retail, where products may experience intermittent demand with frequent zero values. The INGARCH model's ability to account for overdispersion and autocorrelation makes it a robust choice for forecasting in such contexts.

Chu and Zhang (2003) examine the effectiveness of linear regression models in comparison to more complex methodologies for forecasting retail sales. Their study highlights the simplicity and interpretability of linear models, making them attractive for initial modelling phases or in scenarios where resources are constrained. However, they also illustrate that linear models may struggle to capture the non-linear dynamics and interrelationships inherent in retail sales data, particularly when seasonality and product interactions are significant factors. This limitation is especially pertinent in industries like fast fashion, where demand is highly volatile.

To complement these approaches, several additional models have also been explored in recent research. Simple methods such as the Moving Average remain widely used as baseline benchmarks, particularly in early-stage modelling or when historical data is limited (Makridakis et al., 2008). More recently, gradient boosting algorithms like LightGBM and CatBoost have gained traction in the field of retail forecasting. Empirical studies indicate that LightGBM is proficient in capturing trend and seasonality patterns within retail data (Zhou, 2023) and has demonstrated superior performance compared to traditional and neural models in comparative evaluations of large-scale retail sales forecasting (Hobor et al., 2025). Similarly, CatBoost has shown significant potential in the fashion industry, proving to be robust in highly volatile environments, such as women's apparel retail during the Covid-19 crisis (Kizgin et al., 2025). Their inclusion in forecasting studies highlights the increasingly vital role of scalable and accurate machine learning solutions in dynamic retail settings.

These contributions build upon earlier reviews that primarily compared traditional time series models with machine learning approaches. For example, Swaminathan and Venkitasubramony (2024) and Liu et al. (2013) highlighted the benefits of hybrid methods

that merge linear models with machine learning techniques. Furthermore, studies by Hyndman and Athanasopoulos (2018), Teunter et al. (2011) and Chu and Zhang (2003) delved deeper into traditional forecasting methods, such as Holt-Winters smoothing, Croston's method for handling intermittent demand and linear regression models outlining their applications, advantages and drawbacks within retail contexts. Collectively, these works address the integration of external variables, underscore the significance of feature engineering and tackle the challenges of forecasting new products with limited historical data.

These works collectively showcase the increasing diversity of forecasting methods utilised in the retail sector, ranging from traditional statistical models and autoregressive frameworks to count-based distributions and advanced machine learning techniques. Each method presents distinct advantages depending on the data structure, the availability of external information and the particular business context.

Table 1: Summary of Forecasting Techniques and Corresponding Literature in Fashion Retail

Author(s)	Models Used / Focus
Thomassey (2016)	Clustering, Decision Trees, ELM (Hybrid Model)
Martins and Galegale (2022)	ARIMA, Exp. Smoothing, ANN, SVM, Random Forest
Hyndman and Athanasopoulos (2018)	Holt-Winters / Exponential Smoothing
Swaminathan and Venkitasubramony (2024)	Regression, ARIMA, ML, Hybrid
Liu et al. (2013)	ARIMA + ANN / Fuzzy Logic
Teunter et al. (2011)	Croston Method for Intermittent Demand
Freeland and McCabe (2004)	Poisson Autoregressive (PAR), Conditional Median
Kolassa (2016)	Poisson, NB, ZINB, PIT Scoring
Liboschik et al. (2017)	tscount: GLM, INGARCH, ZIP, ZINB
Chu and Zhang (2003)	Linear Regression
Makridakis et al. (2008)	Moving Average
Kizgin et al. (2025)	CatBoost
Zhou (2023)	LightGBM

2.2.3. Daily Sales Forecasting

Many studies have examined demand forecasting in the fashion retail industry, with a significant focus on the medium and long-term predictions that are essential for strategic planning, procurement and production management. However, short-term forecasting,

especially on a daily basis, is equally important as it directly impacts operational decisions such as inventory replenishment, store-level stock distribution and promotional planning. The fashion retail sector is highly volatile, with rapidly shifting trends and short product lifecycles, making the development of reliable daily sales forecasting models a complex challenge (Hyndman & Athanasopoulos, 2018).

Several studies have focused on daily sales forecasting in the retail sector, applying various statistical and machine learning models to enhance prediction accuracy. Liu et al. (2021) introduced a Minimum Description Length Neural Network (MDL-NN) specifically designed to improve daily sales forecasting for SKUs in a multinational fashion retailer. Their approach was particularly effective in handling intermittent sales data, outperforming traditional methods like ARIMA and Exponential Smoothing in capturing demand fluctuations. Likewise, Auppakorn and Phumchusri (2022) investigated daily sales forecasting for retail products with variable pricing. They compared traditional models such as ARIMA with machine learning techniques like XGBoost. Their results revealed that machine learning models were more effective than statistical methods, especially for products whose demand patterns were affected by promotions and external variables.

In addition to traditional statistical and machine learning approaches, hybrid models have been developed to combine the strengths of both methods. Doganis et al. (2006) created a hybrid framework that integrates Radial Basis Function Neural Networks with Genetic Algorithms to forecast daily sales of perishable goods. Their model improved the selection of predictive features and achieved greater accuracy compared to standalone statistical models. These hybrid approaches have become increasingly popular as they allow for the integration of domain knowledge from traditional models while also benefiting from the flexibility of machine learning techniques.

Recent research has emphasised not only the need to improve prediction accuracy, but also the challenges associated with point forecasts in daily sales forecasting. Kolassa (2016) points out that relying exclusively on single-value predictions can be misleading in retail environments, which are often characterised by high variability. He advocates for the use of predictive distributions, which provide a range of potential future sales values along with their associated probabilities. Unlike traditional point forecasts, which estimate only a single expected value, predictive distributions offer a probabilistic perspective. This allows decision-makers to quantify forecast uncertainty and make more informed inventory and replenishment decisions. By considering the full range of possible outcomes, retailers can

more effectively assess the likelihood of stockouts, overstocking and other risks.

The expanding body of research on daily sales forecasting highlights its essential role in managing short-term demand fluctuations within the retail sector. While traditional models continue to be instrumental in capturing structured and seasonal demand patterns, machine learning and hybrid approaches have demonstrated greater effectiveness in addressing irregular sales behaviour, intermittent demand and external influences. As data-driven strategies gain traction in the fashion retail industry, daily sales forecasting is increasingly recognised as a vital tool for optimising inventory management, reducing stockouts and enhancing overall operational efficiency.

2.2.4. Evaluation Metrics for Replenishment Models

The effectiveness of forecasting models in fashion retail is evaluated using various performance metrics to ensure that replenishment strategies are aligned with real-time demand fluctuations and operational objectives. Evaluating the accuracy of forecasts is not only important from a statistical perspective but also as a business necessity, as errors can result in significant financial losses due to stockouts or excess inventory (Hyndman & Athanasopoulos, 2018).

Several key performance indicators (KPIs) are commonly used to evaluate the reliability of forecasting models in the fashion retail sector (Chatfield & Xing, 2019; Christopher, 2022; Fildes et al., 2009; Hyndman & Athanasopoulos, 2018; Kolassa, 2016; Makridakis et al., 2008):

- Mean Absolute Error (MAE): This metric measures the average magnitude of errors without considering their direction, providing a straightforward assessment of prediction accuracy.
- Root Mean Squared Error (RMSE): Similar to MAE, RMSE penalises larger deviations more heavily, making it especially relevant to the fashion industry, where sudden shifts in demand can lead to costly inventory miscalculations.
- Mean Absolute Percentage Error (MAPE): This measure expresses forecast errors as a percentage of actual values, allowing comparability across different product categories.
- Weighted Absolute Percentage Error (WAPE): A variation of MAPE, WAPE accounts for sales volume, helping to prevent the overrepresentation of low-sales

items in the evaluation process.

- **Stockout Rate:** This metric represents the percentage of demand that could not be fulfilled due to insufficient inventory. It indicates how often a forecasting model underestimates demand, leading to stockouts, lost sales and potential customer dissatisfaction.
- **Inventory Turnover Rate:** This metric assesses the frequency at which inventory is sold and restocked over a specific period. It is usually calculated as the ratio of the cost of goods sold to the average inventory level. This provides insight into how efficiently stock is managed in relation to demand.

In fashion retail, where demand is often low or intermittent, the relevance of additional metrics has gained prominence. Traditional percentage-based measures, such as MAPE, can encounter limitations in this context. They tend to become unstable when actual values approach zero and present calculation limitations when actual values are exactly zero. These challenges can skew evaluation results in datasets characterised by sparse or highly variable demand. Consequently, alternative metrics have been proposed to deliver more robust and interpretable evaluations under such circumstances:

- **Symmetric Mean Absolute Percentage Error (SMAPE):** was introduced as a refinement of MAPE, utilising a symmetric formula that restricts the error to a range between 0% and 200%, where 0% indicates perfect accuracy and 200% indicates the worst possible accuracy. However, its formulation can still lead to instability when the actual values are close to zero, which is often the case in datasets characterised by sparse or intermittent demand (Hyndman & Koehler, 2006; Makridakis & Hibon, 2000).
- **SMAPE3:** is a refined version of SMAPE designed to exclude pairs where both the actual and forecast values are zero, thereby preventing misleading inflation or deflation of the metric. This adjustment is particularly advisable for low-demand scenarios where zero values frequently dominate the dataset (Nakashima, 2014). While this modification alleviates some of the extreme behaviour of SMAPE, it can still yield high error scores in contexts with very low sales volumes.
- **Mean Arctangent Absolute Percentage Error (MAAPE):** serves as a modern alternative to MAPE and SMAPE. It effectively avoids issues related to division by small numbers, resulting in stable and interpretable error values, even in scenarios involving intermittent demand. Moreover, since MAAPE is utilised for real-world

discrete targets like sales volumes, several studies suggest rounding continuous forecasts before their calculation. This approach aligns forecasted quantities with actual figures and helps prevent the artificial inflation of error stemming from fractional predictions. Empirical evidence indicates that MAAPE values can vary significantly based on whether rounding is implemented, particularly in low-scale demand contexts, including fashion retail (Kim & Kim, 2016).

While these metrics provide a structured method to evaluate forecasting efficiency, the implementation of sales prediction models can vary widely between companies. Two leading fashion retailers, Zara and H&M, have established advanced forecasting strategies that are specifically tailored to their unique supply chain models. Their approaches illustrate how forecast accuracy impacts replenishment cycles, pricing strategies and overall profitability.

Zara has developed its forecasting and replenishment strategies around agility and rapid response to market trends. The company integrates real-time sales data, artificial intelligence and expert judgement to optimise short-term demand predictions. Unlike traditional inventory systems that rely on large seasonal orders, Zara uses a Just-In-Time (JIT) inventory approach to ensure that new collections reach stores within weeks of production (Bonnefoi, 2012). Zara evaluates the effectiveness of its sales forecasting using various proprietary metrics, including the Y-Metric. This metric assesses the ratio of revenue generated during clearance sales to the original inventory valuation at regular-season prices. By using the Y-Metric, Zara can optimise its markdown strategies while maintaining profitability. Empirical studies have shown forecasting errors in the range of 10% to 14%, which is relatively low given the complexity of demand fluctuations in the fast fashion sector (Carboni Borrásé, 2009). The integration of AI-driven forecasting models has further enhanced Zara's inventory management, leading to a 6% increase in profits from clearance sales. By utilising machine learning techniques and predictive analytics, Zara is continuously improving its forecasting accuracy. This enables a more efficient supply chain, reduces overstock and enhances sell-through rates (Ghemawat, 2003).

H&M has adopted a data-driven approach to sales forecasting, utilising big data analytics and AI-powered models to improve demand projections and optimise inventory management (Reinartz et al., 2019). Unlike Zara's JIT strategy, H&M uses a hybrid approach that balances buffer inventory with demand-driven production. However, despite advances in predictive modelling, H&M has faced forecasting challenges, resulting in \$4.3 billion worth

of unsold inventory in 2018 and missed sales targets during crucial promotional events (Fish, 2023). To evaluate forecasting performance, H&M employs key metrics such as MAPE, WAPE and Sell-Through Rate. MAPE and WAPE assess how much actual sales deviate from predictions across different product categories, while the Sell-Through Rate measures the percentage of inventory sold at full price before markdowns, offering insights into pricing efficiency and stock turnover. These metrics help H&M to minimise inventory misalignment, refine demand predictions and reduce reliance on markdown-driven stock clearance. Although AI-driven forecasting has enhanced predictive accuracy, H&M continues to refine its models to improve short-term demand responsiveness (Reinartz et al., 2019).

These case studies demonstrate how effective forecasting metrics can enhance inventory efficiency, profitability and overall supply chain performance in the fashion retail industry, highlighting the necessity for continuous improvement of predictive models to adapt to market fluctuations.

2.2.5. Cross-Validation in Time Series

When it comes to time series forecasting, it is necessary to use validation methods that maintain the temporal sequence of observations when evaluating predictive models. Traditional cross-validation techniques, like random k-fold (which splits the dataset randomly into k folds), presume independence among data points, consequently disrupting the time-dependent structure of the series. This can result in overly optimistic and misleading assessments of predictive accuracy.

Tashman (2000) was a pioneer in tackling this limitation by introducing rolling-origin evaluation, a method that simulates the real-world forecasting process by progressively expanding the training window as time advances. In this approach, the model is retrained at each step using all available data up to a certain point, followed by testing on the next observation. This technique honours the sequential nature of the data and is particularly recommended for non-stationary series or when recent patterns hold greater significance.

Bergmeir and Benítez (2012) conducted a thorough investigation into the limitations of standard cross-validation techniques in the context of time series data. They highlighted that employing random splits can disrupt temporal dependencies, potentially leading to inaccurate model selection. To address this issue, they introduced blocked cross-validation, a method that segments the dataset into contiguous folds without shuffling, thereby preserving the chronological order. Their experiments demonstrated that this approach yields more

robust and realistic estimates, especially in cases of limited or seasonally structured data.

In a more recent study, Cerqueira et al. (2020) assessed various performance estimation strategies utilising both synthetic and real-world time series data. Their findings indicated that rolling-origin evaluation surpasses other methods, particularly in scenarios characterised by non-stationarity, regime shifts or evolving seasonality, traits commonly observed in retail and fast fashion sales data. They highlighted that validation procedures that fail to consider the order of time often underestimate forecast uncertainty and tend to overfit the training set.

These findings highlight the strong consensus within the forecasting literature on the importance of maintaining temporal dependencies during model evaluation. Both rolling-origin and blocked time series validation have proven to be reliable techniques, providing robust estimates of predictive performance across various domains, especially in scenarios characterised by non-stationarity or limited data availability.

3. A CRISP-DM Based Framework for Sales Forecasting

This chapter presents the practical implementation of the sales forecasting project, structured according to the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology. CRISP-DM is a widely recognised and structured framework that supports the iterative development of data science solutions, placing strong emphasis on business understanding and problem definition (Chapman et al., 2000; Schröder et al., 2021). Due to its flexibility and adaptability, the methodology has been widely adopted in diverse industries and analytical contexts.

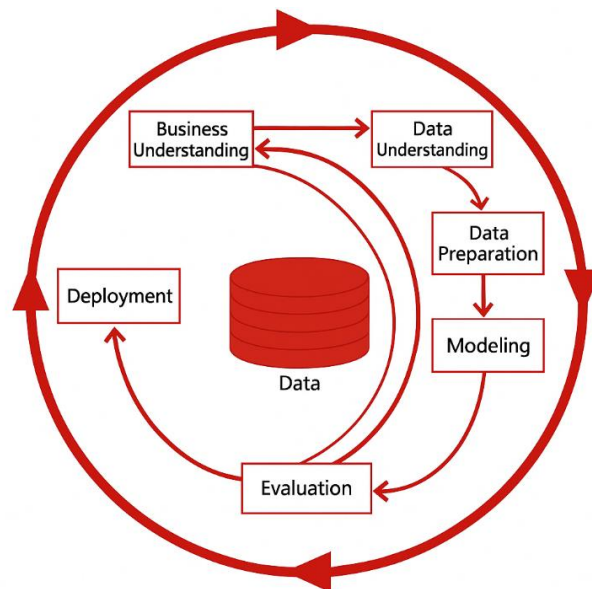


Figure 2: Phases of the CRISP-DM process model. Source: Gallagher, C. V., Bruton, K., & O'Sullivan, D. T. J. (2016). In Y. Tan & Y. Shi (Eds.), *Data Mining and Big Data* (pp. 48–58). Springer.

Given the complexity and dynamism of the fashion retail sector, this methodology provides a robust foundation for exploring and addressing the forecasting challenges posed by intermittent daily sales data. The chapter begins with an overview of the business context and the company's specific needs, followed by a comprehensive analysis of the available data and the preparation steps required for modelling. The final sections outline the modelling phase, including the rationale behind the selection of forecasting algorithms and the evaluation framework used to assess model performance.

3.1. Business Understanding

This Portuguese fashion retail company embarked on its journey with the inauguration of its first store in Porto, initially concentrating on fashion accessories. Over the years, it has evolved into a global brand, boasting more than a thousand stores across 73 countries. Its primary markets include Portugal, Spain, France, Italy, Poland, Switzerland and Belgium, which collectively account for approximately 75% of its total sales. The stores are organised into distribution channels, which group outlets according to factors such as transit times and shipment frequencies. This channel-based strategy allows the company to optimise its supply chain management, ensuring efficient logistics and tailored operations that go beyond a country-level framework.

The company operates under a two-season framework: Spring/Summer (SS) and Fall/Winter (FW). This structure is designed to fulfil the need for continual renewal in the fast fashion industry. The SS collections typically launch in October and extend until August of the following year, while the FW collections commence in April and continue through February. Each season introduces approximately 5000 SKUs, unique codes utilised to track individual product variations, such as different colours or sizes, reflecting the company's strategy to maintain a broad and diverse product assortment.

Products are classified into 20 active categories across five main departments: Bags, Jewellery, Textile Accessories, Non-Textile Accessories and Footwear. As illustrated in Figure 3, handbags emerge as the leading category, representing 28% of total sales for the fiscal year 2023/24. This is followed by Jewellery at 17% and Apparel at 15%. Other noteworthy categories include Wallets at 7%, Footwear at 5% and both Stainless Steel and Night Accessories, each at 4%. This variety in categories underscores the company's dedication to providing a comprehensive range of products that cater to diverse customer preferences and occasions.

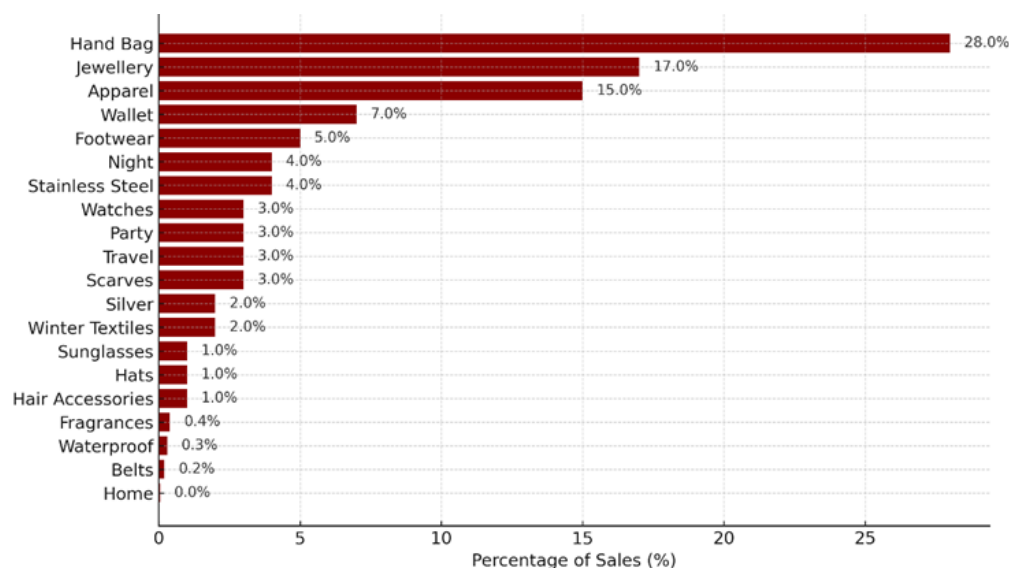


Figure 3: Category Weight – Fiscal Year 2023/24. Source: Internal company data

Reflective of the fast fashion market's dynamics, most of the company's products are refreshed at the start of each new season. However, there are exceptions in the form of "permanent" items, a carefully curated selection of timeless and essential products that remain unchanged by seasonal trends. These items offer a sense of stability and consistency amid a rapidly evolving and ever-changing product line-up.

To meet the rapidly changing demands of customers, the company delivers new products to stores weekly, ensuring that the latest trends and collections are consistently available. The initial stock quantity allocated to each store is determined by expected sales levels, defined by the company's experts. These levels are estimated based on the performance of comparable products during their first days in store in similar periods. From this, a final initial shipment quantity is calculated, taking into account the number of days it is expected to cover, which varies depending on the distribution channel. This allocation process is particularly critical, since the accuracy of forecasts at this stage has a decisive impact on how efficiently stock is distributed. Once a product becomes available in stores, it enters the pool of items eligible for replenishment, depending on whether its expected sales justify further deliveries.

Errors in stock allocation have significant implications for the company. Overestimating demand can lead to excess inventory that must be sold in subsequent periods or through promotions, potentially reducing profit margins. On the other hand, underestimating demand can lead to stockouts, causing lost sales that are difficult to recover

and potentially impacting customer satisfaction. Recognising these risks, the company carefully evaluates the cost implications of both overstocking and stockouts, prioritising a balanced strategy to optimise inventory management and ensure a seamless customer experience.

This carefully designed and flexible operational framework, along with its diverse product range, demonstrates the company's dedication to remaining competitive in the global fashion market. Through a combination of distribution strategies, a diverse product catalogue and structured restocking practices, the company can address customer demand while managing the challenges inherent to the fast fashion industry.

3.2. Data Understanding

As outlined in the previous section, the dataset employed in this study was sourced from the Portuguese fashion retail company under examination. It comprises internal operational data organised into five key components: the central sales dataset, which forms the basis for forecasting, along with four supplementary datasets that detail product characteristics, store metadata, replenishment parameters and the commercial calendar.

The sales dataset provided consists of transactional records from five distinct stores, covering ten different product categories. It captures daily sales transactions from December 2023 to January 2025, with each record representing a unique combination of store, product and date. Two key sales variables are included: one reflects the total recorded sales per day, which may include product returns and can therefore have negative values, while the other represents adjusted sales quantities, which exclude returns. This distinction is essential for understanding customer purchasing behaviour and return patterns, particularly in categories with high return rates, such as apparel and footwear. The selected stores exhibit unique sales patterns and volumes, providing a robust foundation for analysing demand dynamics and supporting the development of forecasting models across various retail contexts. For the visual analysis of sales trends presented in this section, adjusted sales quantities that exclude returns were utilised to ensure a more accurate representation of true demand.

To gain a deeper insight into the dynamics reflected in the sales dataset and to inform the development of precise forecasting models, an initial exploratory analysis was performed. This analysis aimed to identify significant patterns and behaviours at both the store and product levels, emphasising temporal fluctuations, category performance and variability

throughout the product lifecycle.

As illustrated in Figure 4, Store 1013 consistently achieves the highest daily sales volume, followed closely by Store 1016 and Store 1022. All stores experience a notable surge in sales during late 2024, likely coinciding with the Winter season and various promotional activities. These temporal fluctuations emphasise the importance of considering both seasonality and store-specific variations during the modelling phase.

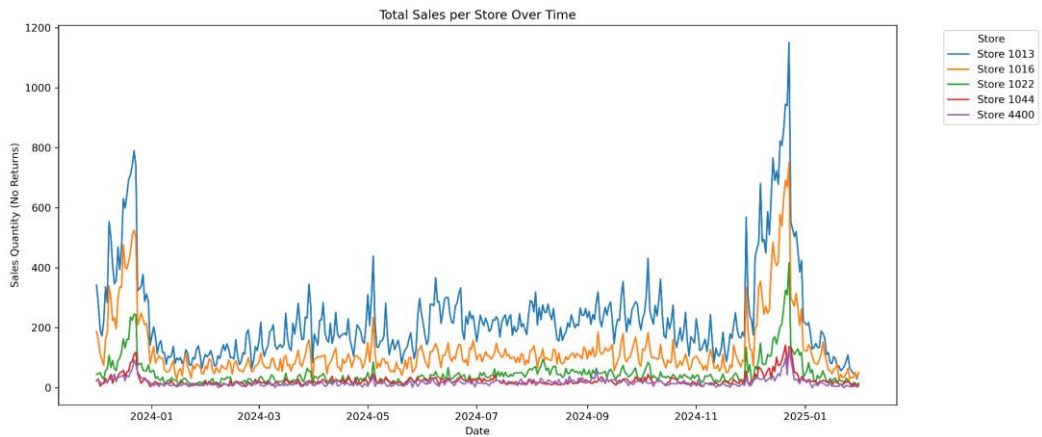


Figure 4: Total Sales per Store over time

In addition to the sales data, a complementary dataset was provided, which included static attributes for each product. Every item is uniquely identified by a product code and characterised by various categorical variables, such as its product category, colour, size and brand family. This dataset also encompasses lifecycle metadata, specifically the launch date of the product and the expected duration of its availability. Furthermore, a binary variable indicates whether a product is classified as permanent, representing a select group of items that remain active across multiple seasons.

Figure 5 illustrates the total sales by product category for Store 1013, which was chosen as a representative example due to its consistently high sales volume. Although this section focuses on a single store, the relative sales dynamics across different product categories remain consistent in the other stores. Consequently, using one store to highlight these trends effectively captures the general patterns observed throughout the retail network.

Jewellery and handbags clearly dominate total sales, followed by apparel and wallets. In contrast, categories like travel, watches and night accessories account for significantly lower volumes. This notable disparity highlights the importance of product categories in demand forecasting, as different segments contribute unevenly to the overall performance

of the store.

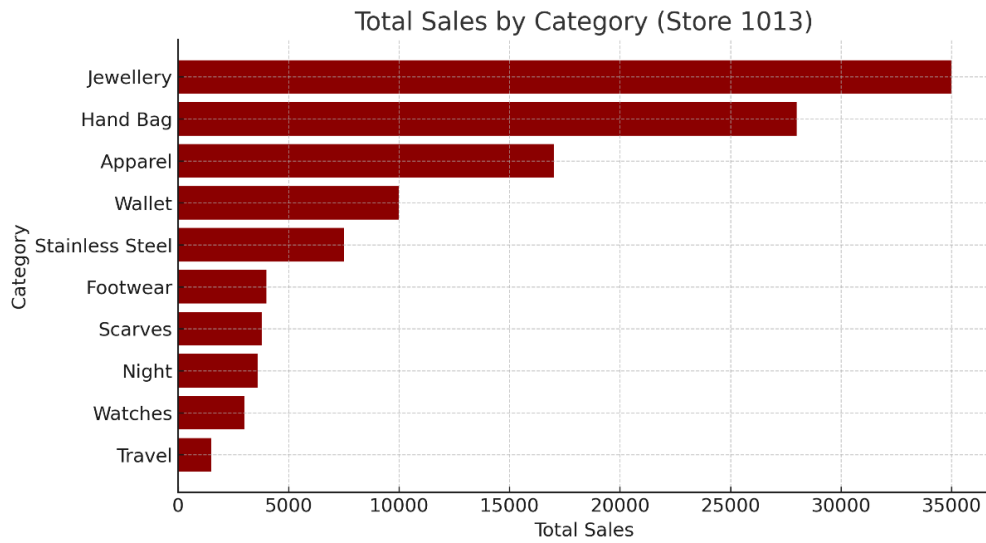


Figure 5: Total Sales by Category (Store 1013)

The data further highlights temporal sales trends across different product categories. Figure 6 illustrates the evolution of daily sales over time for Store 1013, segmented by product type. The figure shows that categories such as jewellery and handbags consistently lead in cumulative sales, while also exhibiting pronounced sales peaks during specific periods. As previously mentioned, these peaks likely coincide with seasonal events or promotional campaigns. Such trends reinforce the importance of incorporating seasonality and promotional effects into the forecasting process to capture demand shifts more accurately.

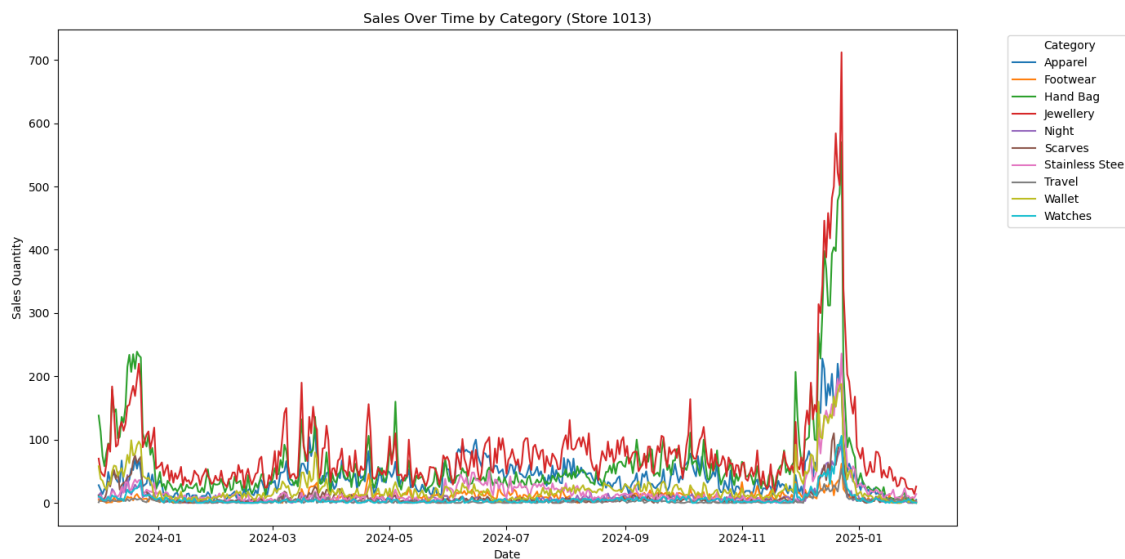


Figure 6: Sales over time by Category (Store 1013)

An important characteristic of the dataset is the variability in product life cycles, indicated by the variable Item Life Cycle Week, which defines the expected number of weeks each product remains in store. In the dataset for Store 1013, the values range from 4 to 24 weeks, with a notable concentration between 8 and 10 weeks. To facilitate analysis, products were categorised into three groups based on this empirical distribution: Short Life Cycle (4 to 6 weeks), Medium Life Cycle (7 to 10 weeks) and Long Life Cycle (12 weeks or more). These thresholds were selected to reflect the most common patterns while ensuring all observed values were included.

As shown in Figure 7, most products fall into the Medium Life Cycle category, with Short Life Cycle items also representing a significant portion. In contrast, Long Life Cycle products are comparatively rare. This distribution reflects the company's fast fashion orientation, which favours high product turnover and frequent assortment changes.

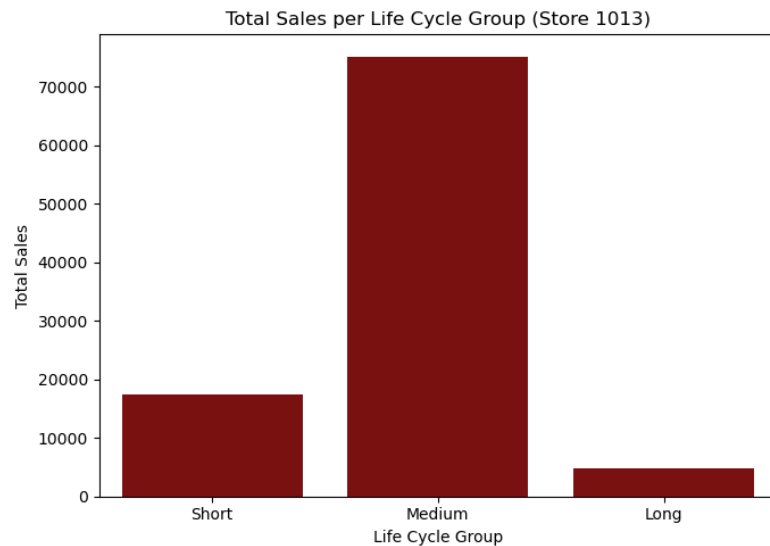


Figure 7: Number of Products per Life Cycle Group

The location dataset offers comprehensive contextual information for each store involved in the analysis. It encompasses variables that describe the geographical setting, like country and district, as well as operational history, including the opening date and structural characteristics like total store area, selling space and warehouse space. Moreover, the dataset includes indicators of commercial classification, differentiating between various store formats (e.g., shopping mall versus street location) and business models. This information supports a deeper understanding of the heterogeneity among stores, particularly in terms of

scale, location and logistical constraints, which is essential for interpreting demand patterns and informing stock allocation and replenishment decisions.

The replenishment dataset comprises essential operational variables related to stock management, forecasting horizons and demand planning. It includes details on replenishment frequency, in-store stock levels, scheduled shipping dates and product lead times. Importantly, it also incorporates forecast values generated by the company's current prediction system, facilitating a comparative analysis between the proposed modelling approach and the existing forecasting methods. This dataset offers valuable insights into how replenishment aligns with anticipated demand and how inventory decisions are executed throughout the supply chain.

The commercial calendar dataset offers vital contextual information regarding promotional activities and store operations throughout the sales period. It encompasses variables that detail the types, start dates and end dates of each campaign, along with daily indicators of store opening status. This information is essential for detecting periods of intensified commercial activity and potential demand peaks. By integrating this dataset into the forecasting process, it becomes possible to consider the influence of marketing initiatives and store availability on sales trends, thereby improving the temporal alignment and accuracy of predictions.

In summary, the datasets provided deliver a comprehensive and multidimensional perspective of the company's operations, integrating static attributes such as product characteristics and store profiles with dynamic variables that reflect sales performance, replenishment behaviour and promotional activity. Collectively, these data sources establish the empirical groundwork for the forthcoming stages of data preparation and modelling. The patterns observed at both the store and category levels, along with the variations in product life cycles, will be further examined during the modelling phase, where they will significantly influence feature engineering and the design of forecasting strategies.

3.3. Data Preparation

The data preparation phase was fundamental to ensure that the modelling process was built upon consistent, meaningful and high-quality data. Given the complexity of daily sales data in a fast fashion retail context, this phase combined business logic, exploratory insights and established data science practices to construct a robust modelling dataset.

Several key steps were undertaken, including data cleaning, time series completion, integration of multiple sources and incremental feature design, all of which contributed to preparing the data for forecasting.

The initial stage of the preparation process consisted of identifying and selecting the most suitable variable to represent demand in the sales dataset. Two candidate variables were considered: one reflecting gross sales figures, which include product returns and can result in negative or zero values on certain days and another capturing the number of units sold, excluding returns. The latter was chosen as the forecasting target, as it provides a clearer and more consistent measure of actual customer purchasing behaviour. This distinction is particularly important in product categories like apparel and footwear, where return rates can be high and volatile, potentially distorting the underlying sales signal. By focusing on adjusted sales figures that exclude returns, the dataset offers a more reliable foundation for modelling true demand patterns across various store-product combinations.

The original sales dataset presented a structural limitation due to its transactional nature: it only captured days on which sales occurred. While this design is typical for operational systems, it poses a considerable constraint for time series forecasting, which relies on complete temporal continuity. To create a consistent and uninterrupted time series for each product-store combination, it was essential to explicitly account for days with no recorded sales. This involved identifying all dates when each product was available for sale, based on its launch date and expected lifecycle and then intersecting this timeframe with the commercial calendar to exclude days when the store was closed (for example, Sundays or public holidays). For all valid dates without a recorded transaction, new entries were generated with the sales quantity set to zero, indicating the product's availability alongside the absence of a sale. This approach ensured that the model would interpret data gaps accurately, learning from periods of inactivity rather than misinterpreting them as missing information. Finally, to avoid creating artificial series, only product-store combinations with at least one recorded transaction were retained.

Before integrating additional datasets into the forecasting table, a comprehensive cleaning and preparation process was conducted. The original product dataset comprised 144 variables, many of which were highly specific to internal documentation, operational classification or system-level tracking. While each of these variables was created with a clear business rationale, not all were considered relevant within the scope of this forecasting study.

As part of the cleaning process, variables exhibiting near-constant values across the

dataset (i.e., with over 99% identical entries) were excluded, as they offered limited variability for modelling purposes. Variables with extensive missing data, exceeding 50%, were also removed, unless they were considered potentially informative from a business perspective, such as those describing material composition, product hierarchy or commercial positioning. However, in the final modelling stage, which focused solely on products from the wallet category, only a subset of these variables was retained and none of them contained missing values. As such, no imputation was required for the categorical variables ultimately included in the forecasting models. Furthermore, a subset of fields generated during an earlier internal project, typically duplicating information already available in cleaner formats, was eliminated to reduce redundancy and improve dataset clarity.

To enhance interpretability throughout the modelling and reporting phases, descriptive fields (e.g., category labels or commercial names) were retained over corresponding technical identifiers or codes. This choice supported clearer communication of insights, particularly when engaging with business stakeholders not directly involved in data engineering or model development.

The location dataset underwent a similar cleaning and evaluation process. Fields with no observations or those that duplicated more reliable information from other sources were removed. Additionally, several attributes related to internal operations, such as employee tracking codes or system-generated timestamps, were excluded due to their limited relevance to the forecasting task. Certain fields describing store status classifications were also discarded after confirming they did not meaningfully distinguish among the selected stores, offering little explanatory value for modelling purposes.

The commercial calendar was streamlined to retain only those variables pertinent to modelling temporal effects, specifically focusing on store openings and promotional activities. Fields associated with campaign naming, creation timestamps and hourly indicators were excluded. This refined version of the dataset maintained only the essential flags and date markers needed for generating calendar-based features and for accurately identifying days when stores were open.

After cleaning and aligning all datasets, the focus shifted to creating the final modelling dataset. Due to the high dimensionality of the data and the exploratory nature of the project, a modular and progressive approach was adopted, which is detailed in the following paragraphs. This strategy facilitated early testing with a simplified set of features and enabled a controlled, iterative evolution of the modelling process.

Although the original sales dataset encompassed ten product categories across five different stores, the modelling phase focused exclusively on a single product category, wallets. This decision was made to ensure computational feasibility while maintaining representativeness. The selected category was among the highest-selling in the dataset, offering sufficient data volume to support robust training and evaluation without introducing excessive computational overhead. Additionally, this choice allowed the modelling pipeline to be tested in a controlled setting, with the potential for future scalability to other categories.

An initial version of the dataset was then constructed using a core set of variables: store identifier, product identifier, transaction date, the demand target (daily sales excluding returns) and lagged sales features that captured recent performance. These included both non-aggregated lag (e.g., the exact sales values from one day prior) and accumulated lags (e.g., rolling sums over the previous 7 and 14 days), allowing the models to learn from both immediate and aggregated temporal patterns.

Although the product category variable was also retained at this stage, it only included the Wallet category, as the modelling exercise was limited to this specific segment due to computational constraints. This category was selected for its business relevance, being among the highest selling and for its manageable size, which enabled faster experimentation. The selected features supported early testing by ensuring low computational complexity while preserving interpretability and alignment with best practices in retail forecasting literature. The initial set of variables used in this simplified dataset is presented in Table 2.

Table 2: Variables in the Initial Modelling Dataset

Variable	Description	Type
Store identifier	Unique code representing each store	Categorical
Product Identifier	Unique code representing each product	Categorical
Date	Transaction date	Date
Sales Quantity (Excl. Returns)	Number of units sold per day, excluding returns	Numerical
Lag 1	Sales of the previous day (non-aggregated)	Numerical
Lag 7	Total sales from the previous 7 days (rolling sum)	Numerical
Lag 14	Total sales from the previous 14 days (rolling sum)	Numerical
Product Category	Product category (only includes <i>Wallet</i> in this case)	Categorical

As the modelling process advanced, the dataset was gradually improved by incorporating additional features derived from the cleaned product and store datasets. These features included attributes like colour, size, lifecycle duration, permanence status and distribution channel, along with calendar-based indicators like campaign periods and seasonal flags. This modular approach facilitated a clear evaluation of the contribution of each variable to forecast accuracy while also controlling the complexity and generalisability of the resulting models.

In the end, the data preparation process ensured that the final dataset was both statistically robust and fully aligned with the business's operational and strategic context. This thoughtful design laid the groundwork for a flexible and scalable forecasting system, capable of adapting to the dynamic nature of the fast fashion retail industry.

3.4. Modelling

The modelling phase is a critical stage in developing a forecasting system, where predictive algorithms are implemented and assessed to tackle the unique challenges of the fashion retail environment. This sector is characterised by rapid product turnover, unpredictable demand and frequent zero sales periods, which often render traditional forecasting methods inadequate. To navigate these complexities, a wide range of techniques was explored, from basic statistical methods to advanced machine learning algorithms and hybrid strategies. Each approach was selected for its potential to achieve a balance between predictive accuracy, computational efficiency and operational practicality.

The modelling pipeline used in this project was designed to be modular and progressive, allowing for the gradual integration of features and different model types. This approach enabled systematic experimentation, starting with simplified setups and gradually moving towards more complex configurations. The primary objectives at each stage was not only to enhance forecasting performance but also to ensure robustness and scalability. This is particularly important in a production context, where the model must generalise effectively to new products and stores without depending on extensive historical data. This chapter outlines the key elements of the modelling phase, beginning with the data splitting strategy and followed by a detailed overview of the modelling techniques explored throughout the project.

3.4.1 Data Splitting and Forecasting Strategy

To create a realistic and scalable evaluation framework, the dataset was divided based on a strictly temporal criterion, reflecting the sequential nature of real-world forecasting tasks. The training set includes all records from December 2023 to November 2024, while the test set encompasses the following period, from December 2024 to January 2025. This division simulates a forward-looking scenario where past data is used to predict future demand, ensuring that there is no information leakage from the test period into the training process.

The decision to adopt a purely date-based split was made in close collaboration with the company, responding to their operational constraints and deployment needs. The company also emphasised the importance of developing a forecasting system capable of generating predictions for all products available at any given time, including newly introduced or returning items. To ensure this, product identifiers were not used as features in the training process, since the model should learn from sales patterns rather than memorising product codes. Consequently, approaches that filter the data by product availability before training were excluded. While these methods are common in academic literature (Salinas et al., 2020; Seeger et al., 2016), they can lead to unrealistic assumptions in dynamic retail environments, where product availability and assortment change frequently.

By focusing exclusively on the temporal sequence of data, the modelling process encourages generalisation based on contextual features such as product attributes, store characteristics and promotional periods. This approach facilitates the model's ability to identify transferable demand patterns and to adapt flexibly to the rapid turnover typical of fast fashion assortments. Additionally, it improves scalability for future expansions, allowing the trained model to forecast demand for any product-store combination, even without prior sales records.

This forecasting strategy combines methodological rigour with practical applicability, establishing a strong, real-time forecasting system that aligns with the company's business objectives.

3.4.2 Traditional Forecasting Methods

To formulate a performance baseline and evaluate the effectiveness of classical forecasting techniques, various traditional statistical models were initially tested. These

included time series smoothing methods like Holt-Winters and Croston's method, as well as autoregressive approaches such as Seasonal Autoregressive Integrated Moving Average (SARIMA) and simple linear regression. While these models provided interpretability and were easy to implement, they consistently underperformed given the dataset's sparse and irregular demand patterns, which are typical of fast fashion sales.

Recognising the limitations of traditional techniques, the analysis shifted towards count-based models specifically designed for discrete and intermittent data. In particular, GLMs using Poisson and NG distributions were examined, along with a ZINB model. These approaches offered a more appropriate probabilistic framework for dealing with data that has many zeros. The ZINB model, in particular, demonstrated enhanced adaptability by combining a binary component to account for zero inflation with a count component that reflects sales magnitude.

Although the theoretical benefits of these count models are significant, none of the traditional methods, when assessed individually, achieved a level of forecasting accuracy that was adequate for operational use. However, insights gained from their performance helped shape the design of subsequent hybrid approaches. In these strategies, components like ZINB and NB regressors were incorporated into more flexible and adaptive forecasting processes.

3.4.3. Machine Learning Models

To address the limitations observed in traditional methods, machine learning algorithms were explored, with a particular focus on tree-based ensemble models. The models tested include Random Forest, XGBoost, LightGBM and CatBoost, all of which are widely acknowledged for their effectiveness in handling structured tabular data and capturing non-linear relationships.

These models were selected for their capability to handle high-dimensional inputs and to automatically model intricate interactions among variables. This is crucial considering the data's nature, which encompasses calendar effects, demand history, product attributes and store-level characteristics.

Although this section does not present specific performance metrics, the results obtained during validation suggested that machine learning models were generally more effective than traditional approaches in capturing demand variability, particularly in the

presence of intermittent or low-volume sales. As such, their integration into hybrid forecasting strategies is further explored in the following section.

Since sales are represented by discrete quantities, the forecasts generated by these models, which inherently produce continuous values, were rounded to the nearest integer before evaluation. This post-processing step ensured alignment with the operational context, where demand is only observable in whole units. It also made the results directly comparable to those from count-based models such as the ZINB and the Negative Binomial GLM, which naturally yield integer predictions.

3.4.4. Hybrid Forecasting Strategies

To address the inherent heterogeneity and sparsity of sales patterns in the fast fashion retail sector, a set of hybrid forecasting models was developed, combining the strengths of multiple techniques according to the characteristics of each product–store time series. The goal of these hybrid approaches is to enhance both flexibility and accuracy by adapting the modelling strategy dynamically based on series length, demand regularity and the occurrence of zero sales.

The first hybrid model tested relied solely on traditional statistical methods. Although traditional approaches generally showed weak performance when used alone, this model was kept for comparison because it utilised a segmented rule-based logic. Specifically, it applied a simple moving average for short time series with fewer than 30 days of data, a ZINB model for highly sparse series, defined as those with 40% or more zero-demand observations and a Negative Binomial GLM for more regular demand patterns. While this model did not achieve competitive results compared to later strategies, it represented an important intermediate step in the modelling process and provided valuable insights into the limitations of statistical methods when facing complex demand dynamics.

Subsequent hybrid models incorporated machine learning components, enhancing their adaptability and predictive performance. One such strategy applied a simple moving average to product-store series shorter than 30 days, a CatBoost regressor to longer series with high sparsity (defined as 40% or more zero-sales observations) and an XGBoost regressor to longer and denser series with more regular demand patterns. This segmentation logic ensured that each model was deployed under the conditions in which it performed best, balancing predictive performance with operational simplicity.

An ensemble-based strategy was explored as an alternative to segmented hybrid

models. This approach integrated predictions from CatBoost, XGBoost, LightGBM and a rolling moving average (with a 7-day window) through a weighted average. Initially, equal weights (0.25 for each model) were assigned to encourage complementarity. To determine a more effective configuration, an exhaustive grid search over the weight combinations was conducted using out-of-sample validation. The optimal configuration allocated 70% of the weight to LightGBM and 30% to the moving average, resulting in improved performance compared to the equal-weight ensemble baseline. This suggests that the combination of the rolling average's stability with LightGBM's predictive strength led to more accurate forecasts during the evaluated test period.

A fourth hybrid strategy improved the segmentation approach by integrating both statistical and machine learning methods in a tiered structure. It utilised moving average forecasts for short time series (less than 30 days), employed a Negative Binomial GLM for highly sparse series (with 70% or more zero sales) and applied XGBoost regressors to all other cases. This flexible design enabled the model to effectively capture both extreme sparsity and regular recurring patterns, allowing it to adapt its behaviour to various demand structures.

Ultimately, two additional hybrid models were created, adhering to the same segmentation principles as Hybrid Model 4. Hybrid Model 5 substituted XGBoost with LightGBM, while Hybrid Model 6 incorporated CatBoost as the boosting component. This design facilitated a direct comparison among the three prominent gradient boosting frameworks, XGBoost, LightGBM and CatBoost, within an identical hybrid framework, offering valuable insights into their relative performance when paired with statistical methods.

All of these hybrid strategies emphasise the need to tailor forecasting methods to the unique characteristics of the data, especially in environments characterised by short life cycles and fluctuating sales patterns. Chapter 4 will provide a detailed analysis of their comparative performance.

Table 3: Summary of Hybrid Forecasting Models

Model	Description
Model 1 – Traditional Hybrid	• MA for <30 days • ZINB for high zero rate • NB GLM for regular series
Model 2 – Rule-Based ML Hybrid	• MA for <30 days • CatBoost for $\geq 40\%$ zeros

	• XGBoost otherwise
Model 3 – Weighted Ensemble	• Combines CatBoost, XGBoost, LightGBM, MA • Initial weights: 0.25 for each one. • Best combination: 0.70 LightGBM and 0.30 MA
Model 4 – Mixed Hybrid (MA + NB GLM + XGBoost)	• MA for <30 days • NB GLM for $\geq 70\%$ zeros • XGBoost otherwise
Model 5 – Mixed Hybrid (MA + NB GLM + LightGBM)	• MA for <30 days • NB GLM for $\geq 70\%$ zeros • LightGBM otherwise
Model 6 – Mixed Hybrid (MA + NB GLM + CatBoost)	• MA for <30 days • NB GLM for $\geq 70\%$ zeros • CatBoost otherwise

3.4.5. Feature Engineering

Developing an effective forecasting model involves not only selecting appropriate algorithms but also carefully choosing features that can accurately capture relevant sales dynamics. In this project, as previously mentioned, feature engineering was carried out incrementally, following a modular approach that aligns with the exploratory nature of the research. This section outlines the primary strategies used for feature selection, which include adding informative variables and excluding features that were considered unsuitable for the modelling task.

To enhance the dataset and improve predictive accuracy, several new variables were gradually introduced. The first was a binary indicator that marked whether a specific day was part of a promotional campaign. This variable aimed to account for the potential increase in sales often seen during marketing events. Its incorporation allowed the forecasting models to distinguish between regular sales patterns and those driven by campaigns, which is a crucial consideration in fast fashion retail settings.

Temporal dependencies were further examined through the addition of lag features. In addition to the previously implemented lags at 1, 7 and 14 days, capturing immediate, weekly and biweekly effects, three additional variables were created to reflect more nuanced short-term dynamics.

The first is a two-day lag of individual product sales, which measures the non-aggregated value observed exactly two days prior. This design choice aims to capture short-term purchasing patterns, such as alternating-day buying behaviours or delayed responses to previous demand, without smoothing the signal through accumulation.

The second is a one-day lag of category-level sales within the same store. This lag aggregates the sales of all products within a specified category, reflecting broader demand trends that may influence related items.

The third is a one-day lag of average store-level sales, calculated across all products in a given store on the previous day. This feature is designed to capture general fluctuations in store performance and customer traffic.

Lag features were introduced to integrate multi-level temporal signals into forecasting models, improving their ability to recognise short-term demand patterns at various levels of detail. Their design was guided by practical insights from the company, which emphasised the importance of both individual and aggregated sales dynamics in operations.

To capture recurrent temporal patterns, two additional features were created: day of the week and month. These variables help account for weekly and monthly sales cycles and seasonal effects, which are particularly relevant in retail contexts. Their inclusion allows the models to better adjust forecasts based on routine consumer behaviour and calendar-driven fluctuations.

Additionally, a selection of product-level attributes was integrated into the main dataset to include information about product positioning, lifecycle and merchandising strategy. The following variables were selected based on their potential impact on customer preferences and sales performance:

- Price-related features, including the product's base price;
- Lifecycle indicators, like the number of weeks a product is expected to remain active;
- Binary classification of permanent products, distinguishing stable, all-year items from seasonal collections;
- Composition and colour-related features, which may reflect consumer preferences for certain materials or aesthetics;
- Collection or campaign theme indicators, providing context for trend-driven products;
- Stock-related information, including the number of units initially made available per product.

Categorical variables were converted into numerical representations to ensure compatibility with the machine learning algorithms used. Various encoding strategies were implemented based on the specific characteristics and cardinality of each feature.

For low-cardinality variables, such as the day of the week, month and colour type, one-

hot encoding is utilised to create binary indicator variables for each category. This approach is especially well-suited for cyclic temporal features and small categorical domains, as it maintains the lack of ordinal relationships between categories while allowing models to capture effects that are specific to each category without imposing any artificial ordering (Géron, 2022; James et al., 2013; Kuhn & Johnson, 2013).

In the case of high-cardinality variables like product composition and collection theme, label encoding was favoured to prevent the dimensionality expansion and sparsity that often accompany one-hot encoding in large-scale retail datasets (Hancock & Khoshgoftaar, 2020; James et al., 2013). Each unique category was assigned an integer identifier, ensuring computational efficiency while maintaining distinctions between categories (Prokhorenkova et al., 2018).

In contrast, categorical encoding was purposefully excluded from the hybrid models. Empirical tests indicated that utilising one-hot or label encoding in this context led to increased sparsity and heightened risks of overfitting, without enhancing forecasting accuracy. Consequently, all variables dependent on categorical transformations, including product composition, collection theme and colour type, were omitted from the hybrid models. However, temporal categorical features were retained: the day of the week and the month were included in their integer forms, as this representation consistently yielded stronger results in the hybrid configuration.

After conducting extensive exploratory analysis and consulting with the business, the variables listed in Table 4 were selected to ensure that only features with significant explanatory power were included in the modelling phase. Some of the variables present in the final dataset were deliberately excluded from the set of predictors to avoid potential information leakage and to enhance model generalisation. Specifically, unique identifiers such as product and store codes were retained in the dataset for reporting and evaluation purposes, enabling forecasts to be linked back to the corresponding store–product combinations; however, they were not utilised during model training to prevent overfitting to individual series. Likewise, the product category variable was retained in the dataset for completeness but was excluded from modelling, given that the study focuses solely on a single product category (wallets), rendering it unnecessary as a predictive feature.

Table 4: Variables in the final Modelling Dataset

Variable	Description	Type	Stage Added	Usage Across Models
Store identifier	Unique code representing each store	Categorical	Initial dataset	Excluded
Product Identifier	Unique code representing each product	Categorical	Initial dataset	Excluded
Product Category	Product category (only includes <i>Wallet</i> in this case)	Categorical	Initial dataset	Excluded
Transaction Date	Date of the transaction (daily granularity)	Date	Initial dataset	All models
Daily Sales (Excl. Returns)	Target variable: units sold, excluding returns	Numerical	Initial dataset	All models (target)
Lag – 1 Day	Sales on the previous day	Numerical	Initial dataset	ML + Hybrid
Rolling Sum – 7 Days	Total sales over the past 7 days (excluding the current day)	Numerical	Initial dataset	ML + Hybrid
Rolling Sum – 14 Days	Total sales over the past 14 days (excluding the current day)	Numerical	Initial dataset	ML + Hybrid
Campaign Day	Flag indicating if a campaign was active	Binary	Feature enrichment	ML + Hybrid
Lag – 2 Days	Sales two days before	Numerical	Feature enrichment	ML + Hybrid
Category Lag – 1 Day	Category-level sales on the previous day	Numerical	Feature enrichment	ML + Hybrid
Store Lag – 1 Day	Store-level sales on the previous day	Numerical	Feature enrichment	ML + Hybrid
Day of the week	Day of the week (0 = Monday, ..., 6 = Sunday)	Categorical	Feature enrichment	ML + Hybrid
Month	Month of the year (1 to 12)	Categorical	Feature enrichment	ML + Hybrid
Product Price	Base price including VAT	Numerical	Feature enrichment	ML + Hybrid
Permanent Product	Indicates if the product belongs to the permanent collection	Binary	Feature enrichment	ML + Hybrid
Material Composition	Material composition description	Categorical	Feature enrichment	ML only
Colour Type	Type of colour (e.g., basic, fashion)	Categorical	Feature enrichment	ML only
Collection Theme	Product theme or visual concept	Categorical	Feature enrichment	ML only

Product Life Cycle	Weeks the item was available in the store	Numerical	Feature enrichment	ML + Hybrid
Initial Units Available	Initial number of units introduced	Numerical	Feature enrichment	ML + Hybrid
Store sales target	The estimated revenue or sales target assigned to the store for one year	Numerical	Feature enrichment	ML + Hybrid
Average Product Sales	Average sales per product in the store	Numerical	Feature enrichment	ML + Hybrid
Store Sales Std. Deviation	Standard deviation of product sales in the store	Numerical	Feature enrichment	ML + Hybrid
Store Sales Percentile	Product's percentile in the store's sales distribution	Numerical	Feature enrichment	ML + Hybrid

This feature selection strategy laid the foundation for the construction of robust and scalable forecasting models, capable of capturing relevant demand signals while avoiding excessive complexity or overfitting.

3.4.6. Tuning

To explore the possibility of improving forecasting accuracy via hyperparameter optimisation (adjustment of external configuration values that control the learning process of a model, like tree depth and learning rate), a dedicated tuning phase was established for the primary machine learning models: LightGBM, XGBoost and CatBoost. Each model was individually fine-tuned using strategies supported by existing literature, tailored to address the specific challenges of forecasting sparse and intermittent sales data within the fast fashion sector. The tuning approach was not identical across models, as the methods were selected according to the optimisation tools most established for each algorithm and their documented effectiveness in comparable forecasting contexts. Throughout the process, all procedures adhered to the temporal structure of the data by employing time-aware cross-validation.

For the LightGBM model, three tuning strategies were explored. Initially, RandomizedSearchCV was utilised to efficiently sample from a broad hyperparameter space, in line with the methodology proposed by Bergstra and Bengio (2012). The hyperparameter grid used and the best configuration identified are shown in Table 5.

Table 5: *LightGBM – RandomizedSearchCV Parameters*

Parameter	Values Tested	Selected Value
num_leaves	[15, 31, 63]	31
max_depth	[-1, 5, 10]	-1
learning_rate	[0.01, 0.05, 0.1]	0.01
n_estimators	[100, 200, 300]	100
min_child_samples	[10, 20]	20
subsample	[0.7, 0.9, 1.0]	0.9
colsample_bytree	[0.7, 0.9, 1.0]	0.7
reg_alpha	[0.0, 0.1]	0.1
reg_lambda	[0.0, 0.1]	0.1

Next, early stopping was tested by creating a validation set from the latter portion of the training period, aiming to prevent overfitting by halting the boosting iterations at the optimal moment (Ke et al., 2017; Prechelt, 2002). Lastly, a grid search approach was adopted to systematically evaluate targeted combinations of essential parameters, as shown in Table 6, including the learning rate and maximum tree depth (Géron, 2022).

Table 6: *LightGBM – Grid Search Parameters*

Parameter	Values Tested	Selected Value
max_depth	[-1, 5, 10]	-1
num_leaves	[15, 31]	15
learning_rate	[0.01, 0.05, 0.1]	0.01
min_child_samples	[10, 20]	10

For XGBoost, two tuning techniques were employed. A grid search was first conducted, targeting essential parameters such as max_depth, min_child_weight and learning_rate, as presented in Table 7.

Table 7: *XGBoost – Grid Search Parameters*

Parameter	Values Tested	Selected Value
max_depth	[0, 5, 10]	5
learning_rate	[0.01, 0.05, 0.1]	0.01
n_estimators	[100, 200]	100
min_child_weight	[10, 20]	10
subsample	[0.8, 1.0]	1.0
colsample_bytree	[0.7, 1.0]	0.7

This was subsequently followed by Bayesian optimisation, utilising the Optuna framework, which adaptively proposed promising hyperparameter values by modelling the

relationship between previous configurations and their performance through a Tree-structured Parzen Estimator (Akiba et al., 2019). The best trial identified through this method is summarised in Table 8.

Table 8: XGBoost – Bayesian Optimisation (Optuna)

Parameter	Value Range	Best Value Found
max_depth	[3, 10]	4
learning_rate	[0.01, 0.3]	0.01038
n_estimators	[100, 300]	127
min_child_weight	[1, 10]	10
subsample	[0.6, 1.0]	0.82858
colsample_bytree	[0.6, 1.0]	0.96669
reg_alpha	[0.0, 1.0]	0.08738
reg_lambda	[0.0, 1.0]	0.09895

The CatBoost model was tuned utilising a randomised hyperparameter search approach, allowing for comprehensive yet computationally manageable exploration of various configurations. This methodology was selected given its suitability for handling high-dimensional parameter spaces and categorical features, which are inherent to the model architecture (Prokhorenkova et al., 2018). To guide the tuning process, a set of relevant hyperparameters was defined, covering aspects like tree depth, learning rate and regularisation. The tested values and the optimal configuration obtained are summarised in Table 9.

Table 9: CatBoost – Randomised Search Parameters

Parameter	Tested Values	Selected Value
learning_rate	[0.01, 0.05, 0.1]	0.1
Depth	[4, 5, 6, 7, 8]	8
iterations	[100, 200, 300]	100
l2_leaf_reg	[1, 3, 5, 7, 9]	5
bagging_temperature	[0.0, 0.5, 1.0]	1.0

Hyperparameter tuning was not implemented for the hybrid models created in this project. These strategies amalgamate forecasting techniques utilising predefined rules based on product and demand characteristics and any adjustments to the machine learning component would only impact a portion of the dataset. Similarly, traditional statistical models, such as Moving Averages or GLMs, were not subjected to tuning, as they depend on analytical estimation methods that do not necessitate manual parameter selection.

3.4.7. Cross-Validation Approach

To evaluate the generalisability and robustness of the forecasting models, a time series-specific cross-validation strategy was implemented. Given the sequential nature of sales data, traditional random k-fold techniques were deemed inappropriate, as they would violate the temporal structure and introduce data leakage. Instead, an expanding window walk-forward validation approach was adopted within the testing period.

In this method, the model is first trained using all available data up to a specified point in time and validated on the following time window. After each iteration, the training set is expanded to include the most recent validation window and the process is repeated. This structure reflects real-world forecasting environments, where models are regularly updated as new sales data becomes available, ensuring that historical data is used to predict future outcomes.

To simulate the forecasting process within the operational deployment timeframe, cross-validation was applied exclusively to the test period, encompassing December 2024 and January 2025. A rolling 7-day forecast horizon was adopted at each step, using an expanding training window that started at the beginning of the test interval. This approach mimics a realistic scenario in which weekly forecasts are generated as new sales data becomes available. Updating the training set week by week allows the model to progressively incorporate information from the most recent period, thereby improving its ability to capture short-term patterns and respond to evolving demand dynamics. The figure below illustrates the first three iterations of this walk-forward validation strategy applied within the test set.



Figure 8: Rolling Forecast Validation with Expanding Training Window (Illustrative Example)

By maintaining the chronological order of events, this validation method allowed for a realistic assessment of the model's performance across various seasonal phases, product

life cycles and promotional periods. The consistency of the results across different folds reinforced the models' stability and reliability. This justified the decision to retrain the models using the complete training period from January to November 2024 before conducting the final evaluation on the test set.

4. Results and Evaluation

This chapter offers a thorough evaluation of the forecasting models developed in Chapter 3, emphasising their predictive performance and practical applicability within the fast fashion retail sector. The assessment integrates three complementary perspectives: statistical accuracy metrics, graphical analysis of forecast errors and a comparative evaluation against the company's existing forecasting methods. As previously noted, the evaluation utilised a representative dataset that encompasses five physical stores and one product category.

4.1. Forecasting Results

The results presented in this section pertain to the final modelling stage, utilising the dataset obtained after feature engineering, with the variables summarised in Table 4 of the preceding chapter. This evaluation focuses on the period from December 2023 to January 2025, encompassing five physical stores and a single product category (wallet). The training set includes data from December 2023 to November 2024, while the test set consists of data from December 2024 and January 2025. This division was designed to replicate a realistic short-term forecasting scenario, in alignment with business operations.

Since sales represent discrete quantities, all forecasts were rounded to the nearest integer. This adjustment was applied uniformly across all models to maintain consistency in methodologies and comparability of results. It also reflects the operational reality of retail demand, which can only be measured in whole units.

After assessing a diverse array of statistical, machine learning and hybrid methods, six models were chosen for final comparison due to their consistent performance and robustness observed during the validation phase. These models include the standalone XGBoost, LightGBM and CatBoost, along with three hybrid configurations: Hybrid Model 4, which integrates Moving Average, Negative Binomial GLM and XGBoost; Hybrid Model 5, which follows the same logic but replaces XGBoost with LightGBM; and Hybrid Model 6, designed with the same segmentation structure but substituting the boosting component with CatBoost.

The evaluation of forecasting performance relied on three complementary error metrics: MAE, RMSE and MAAPE. These metrics were selected to reflect statistical

performance in the presence of low-volume and highly intermittent demand patterns. Such characteristics are common in fast fashion retail, where traditional percentage-based metrics often become unstable or undefined due to zero or near-zero values in the target variable.

Initially, SMAPE3 was considered, as it has been suggested in the literature as an alternative to MAPE for intermittent demand scenarios. However, in this dataset, sales values were often very low, mainly ranging from 0 to 3 units. Consequently, even minor forecast errors resulted in disproportionately high SMAPE3 values. For instance, a one-unit error with a denominator of 2 yields 100%, while with a denominator of 1, it reaches 200%. This behaviour distorted the interpretation of performance, rendering the metric unsuitable for the current analysis. Therefore, SMAPE3 was excluded from the evaluation phase and MAAPE was retained as a more stable percentage-based measure under these conditions.

The use of MAE and RMSE provides insight into the average magnitude of forecast errors, with RMSE placing greater emphasis on larger deviations. Meanwhile, MAAPE offers a robust alternative to MAPE, better suited to the nature of the data analysed in this study.

The mathematical formulations of the selected metrics are presented below:

$$MAE = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}$$

$$MAAPE = \frac{1}{n} \sum_{t=1}^n \arctan\left(\frac{|\hat{y}_t - y_t|}{|y_t| + \epsilon}\right)$$

where y_t denotes the actual value at time t , $\hat{y}_t$ the forecasted value at time t , n the number of observations and ϵ is a small constant added to avoid division by zero.

Unlike MAE and RMSE, which are unbounded, MAAPE is constrained to the interval $[0, \pi/2]$, roughly corresponding to $[0, 1.57]$. When expressed in percentage terms, this corresponds to a maximum of $\pi/2 \times 100 \approx 157\%$. This characteristic ensures that even in extreme scenarios, such as forecasting zero when the actual value is one, the error is capped. As a result, MAAPE serves as a stable and interpretable metric, particularly suited for situations involving low-volume and intermittent demand.

Before comparing the various forecasting models, a preliminary assessment was conducted to determine whether hyperparameter tuning could improve predictive

performance. As outlined in Section 3.4.6, several tuning strategies were employed for LightGBM, XGBoost and CatBoost, including grid search, randomised search and Bayesian optimisation (Optuna). The tuning was performed on the final dataset described in Table 4 (Section 3.4.5, Feature Engineering), using the designated training and validation partitions. The results of these tuning experiments, summarised in Table 10, revealed mixed results when compared to the baseline configurations.

Table 10: Summary of Tuning Results (MAE and RMSE)

XGBoost				
	Baseline	Grid search	Bayesian Optimisation with Optuna	
MAE	0.1456	0.1385	0.1391	
RMSE	0.3619	0.4703	0.4666	
MAAPE	11.68 %	8.68%	9.02%	
LightGBM				
	Baseline	RandomizedSearchCV	Early stopping	Grid search
MAE	0.1410	0.1386	0.1428	0.1394
RMSE	0.3617	0.4693	0.4653	0.4707
MAAPE	10.59%	8.77%	9.99%	8.86%
CatBoost				
	Baseline	RandomizedSearchCV		
MAE	0.1354	0.1420		
RMSE	0.3542	0.4595		
MAAPE	10.20%	10.50%		

For XGBoost and LightGBM, hyperparameter tuning led to a reduction in MAE and MAAPE, indicating a decrease in average forecasting errors. However, this improvement came at the cost of increased RMSE values. This trade-off suggests that while the tuned models achieved greater accuracy on average, they also became more susceptible to extreme errors, which is particularly concerning in the context of intermittent and highly variable demand. In contrast, CatBoost performed even less favourably: tuning not only failed to improve accuracy but consistently worsened performance across all metrics.

The findings indicate that, particularly in the context of short time series,

intermittent demand and significant variability, baseline configurations achieve a more stable equilibrium between accuracy and robustness compared to heavily parameterised alternatives. Therefore, the subsequent analysis will concentrate solely on the baseline versions of the models, as they consistently yielded the most reliable results. The average performance of the six selected models, assessed using the three complementary error metrics, is shown in Table 11 (the best value for each performance measure is shown in bold). The results were obtained by first computing errors at the store-product level and subsequently averaging them across all combinations so that each store-product pair contributed equally to the overall evaluation.

Table 11: Accuracy measures, MAE, RMSE and MAAPE for XGBoost, LightGBM, CatBoost, Hybrid Model 4, Hybrid Model 5 and Hybrid Model 6

Metric	XGBoost	LightGBM	CatBoost	Hybrid Model 4	Hybrid Model 5	Hybrid Model 6
MAE	0.1456	0.1410	0.1354	0.1508	0.1508	0.1517
RMSE	0.3619	0.3617	0.3542	0.3508	0.3508	0.3526
MAAPE	11.68 %	10.59%	10.20%	13.76 %	13.76 %	13.90%

Among the standalone boosting algorithms, CatBoost emerged as the top-performing model, attaining the lowest error values for MAE and MAAPE. It achieved an MAE of 0.1354, an RMSE of 0.3542 and an MAAPE of 10.20%. LightGBM followed closely behind, with a MAE of 0.1410, RMSE of 0.3617 and MAAPE of 10.59%. XGBoost ranked third, displaying slightly higher error values with an MAE of 0.1456, RMSE of 0.3619 and MAAPE of 11.68%. The performance difference between LightGBM and XGBoost is minimal, indicating that both remain competitive alternatives, although they are less accurate than CatBoost.

In contrast, the hybrid models fell short when compared to all standalone boosting approaches. Hybrid Models 4 and 5 yielded identical results, with an MAE of 0.1508, an RMSE of 0.3508 and an MAAPE of 13.76%. Hybrid Model 6, which incorporated CatBoost within the same segmentation logic, performed slightly worse than the other two hybrid configurations, with higher MAE (0.1517), RMSE (0.3526) and MAAPE (13.90%). While the RMSE values of all three hybrid configurations remain comparable to those achieved by the standalone models, their consistently higher MAE and MAAPE indicate that the integration of traditional statistical methods did not enhance predictive accuracy in this context.

To complement the global evaluation, the analysis was further stratified according to the product life cycle, following the categorisation previously established in the data analysis phase. Products were grouped into three categories: Short Life Cycle (4–6 weeks), Medium Life Cycle (7–10 weeks) and Long Life Cycle (12 weeks or more). This segmentation was introduced to investigate whether forecasting performance varied depending on product lifespan. However, within the defined test period (December 2024 to January 2025), no products with a Long Life Cycle were active, which limited the evaluation to the Short and Medium groups.

It is essential to note that although the dataset distinguished between permanent and non-permanent products, a separate analysis of this characteristic was not conducted. The representation of permanent products within the dataset was significantly less than that of non-permanent items, which also reflects the overall composition of the company’s product portfolio. This discrepancy, coupled with the limited temporal coverage of the study period, further restricted the ability to draw meaningful conclusions from an individual analysis. As a result, the findings are presented without differentiation by permanence, highlighting life cycle length as a more informative analytical dimension. The results of this stratified analysis are presented in Table 12 (for each evaluation metric, the best result is indicated in bold).

Table 12: Forecasting performance by product life-cycle group

	Short Life Cycle			Medium Life Cycle		
	MAE	RMSE	MAAPE	MAE	RMSE	MAAPE
XGBoost	0.2192	0.5480	22.17%	0.1378	0.4537	9.53%
LightGBM	0.1941	0.5291	17.70%	0.1390	0.4614	9.28%
CatBoost	0.1649	0.4963	13.03%	0.1392	0.4495	10.16%
Hybrid Model 4	0.1807	0.5107	16.00%	0.1631	0.4648	15.27%
Hybrid Model 5	0.1807	0.5107	16.00%	0.1636	0.4656	15.31%
Hybrid Model 6	0.1807	0.5107	16.00%	0.1638	0.4658	15.34%

The stratified evaluation based on the product life cycle reveals a clear distinction in forecasting performance. Products with a short life cycle, ranging from 4 to 6 weeks, consistently exhibited higher error values across all models, with MAAPE percentages between 13% and 22%. This underscores the inherent challenges of forecasting items with limited historical data, a common issue in the fast fashion industry characterised by rapid

turnover. For this group, CatBoost achieved the best results, particularly with a MAAPE of 13.03%. Conversely, products with a medium life cycle of 7 to 10 weeks exhibited significantly lower error levels, with MAAPE values ranging from 9% to 15%. This suggests that extended product availability contributes to more stable and accurate forecasts. In this case, XGBoost and LightGBM stood out as the most accurate models.

In addition to the numerical evaluation, the distribution of signed forecast errors ($\hat{y}_t - y_t$) was examined for each of the five models. The resulting histogram offers a detailed perspective on the frequency and direction of prediction errors, facilitating a qualitative assessment of the bias and dispersion of each model.

Figure 9 presents the signed error distributions for both the standalone and hybrid models. This visualisation offers a more nuanced understanding of each model's performance, going beyond mere average error metrics to capture not only the magnitude of errors but also their directionality and frequency. By analysing this histogram, consistent patterns can be identified, like tendencies for underestimation or overestimation, as well as assessing how predictions cluster around the correct values. In addition to the graphical representation, Table 13 provides a numerical summary of the signed forecast errors across models.

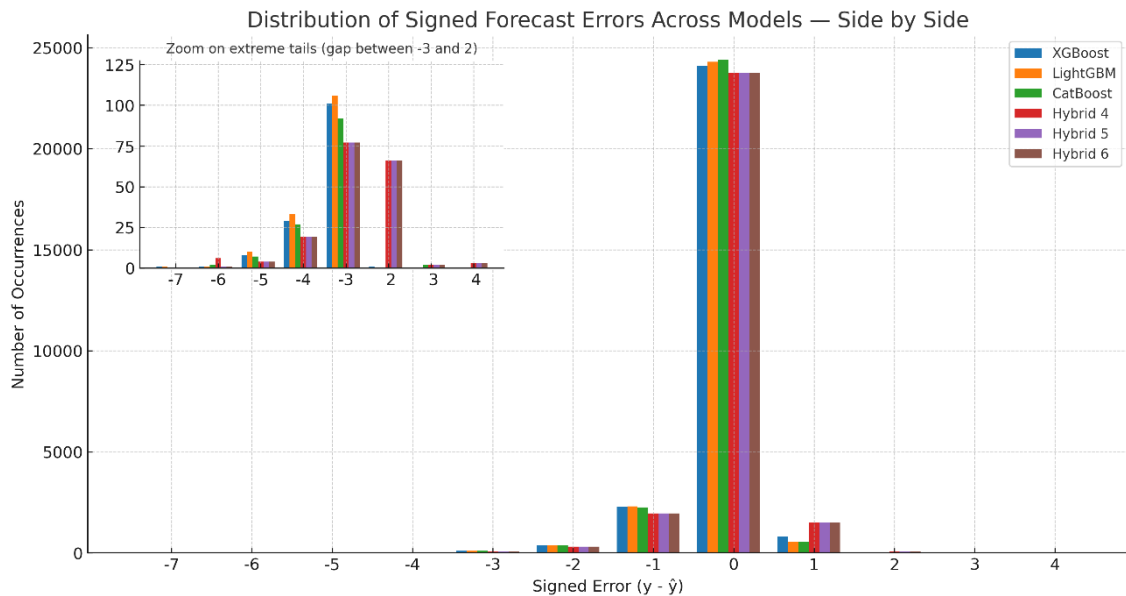


Figure 9: Distribution of signed forecast errors across models. The inset highlights the error tails to better visualise low-frequency occurrences.

Table 13: Comparative Summary of Signed Forecast Errors Across Models

Error	XGBoost	LightGBM	CatBoost	Hybrid Model 4	Hybrid Model 5	Hybrid Model 6
-7	1	1	-	-	-	-
-6	1	1	2	6	1	1
-5	8	10	7	4	4	4
-4	29	33	27	19	19	19
-3	101	106	92	77	77	77
-2	366	384	370	311	311	311
-1	2263	2306	2255	1946	1946	1946
0	24111	24296	24403	23763	23763	23759
1	810	554	535	1499	1499	1503
2	1	-	-	66	66	66
3	-	-	-	2	2	2
4	-	-	-	3	3	3

CatBoost achieved the highest number of exact matches, recording 24403 cases, narrowly surpassing LightGBM (24296) and XGBoost (24111). All three boosting models demonstrated sharply concentrated error distributions, with the majority of deviations falling within the range of ± 1 . Notably, the counts of -1 errors were quite similar across the models, with XGBoost at 2263, LightGBM at 2306 and CatBoost at 2255. In contrast, the +1 overestimations varied, ranging from 535 in CatBoost to 810 in XGBoost. Larger deviations beyond ± 3 were infrequent, never exceeding a few dozen cases, highlighting the overall stability of these boosting methods.

The hybrid models (Hybrid 4, Hybrid 5 and Hybrid 6) exhibited a distinct error structure. Although they achieved a notable number of exact matches (23763 for Hybrids 4 and 5 and 23759 for Hybrid 6), their overall performance was compromised by a significantly higher incidence of near-miss errors. All three models recorded 1946 instances of underestimation by -1, while overestimation by +1 amounted to 1499 cases in Hybrids 4 and 5 and 1503 in Hybrid 6. In comparison to the standalone models, the hybrid configurations displayed greater variability at more extreme levels, including deviations of -6, -7 and +2 to +4, which were primarily absent in XGBoost, LightGBM and CatBoost. This suggests that, although the hybrid methods frequently provided close approximations of demand, their

forecasts were generally more dispersed, resulting in occasional significant deviations from the actual values.

In summary, CatBoost, LightGBM and XGBoost have demonstrated their reliability through highly concentrated error distributions, minimal instances of large deviations and a significant number of exact matches. Notably, CatBoost exhibited the most precise alignment with actual values. In contrast, the hybrid methods, though still competitive, displayed wider spreads around the true values and sporadic extreme deviations, which reduced their overall reliability.

The evaluation was subsequently followed by a walk-forward cross-validation procedure to assess the models' stability across various forecast horizons. In contrast to the single hold-out test, which measures performance over a fixed period, this approach generates forecasts progressively, week by week, providing a more reliable indication of model performance in operational settings. The global results reported in Table 14 (the best score for each performance metric is marked in bold) correspond to the average errors across all weeks, whereas Figure 10 to Figure 12 illustrate the weekly evolution of performance, thereby offering a more detailed view of model stability in operational settings.

Table 14: Summary of Average Cross-Validation Results (MAE, RMSE and MAAPE)

Metric	XGBoost	LightGBM	CatBoost	Hybrid Model 4	Hybrid Model 5	Hybrid Model 6
MAE	0.1241	0.1274	0.1218	0.1251	0.1251	0.1251
RMSE	0.3814	0.3827	0.3809	0.3852	0.3852	0.3852
MAAPE	9.25%	10.0%	8.67%	9.08%	9.08%	9.08%

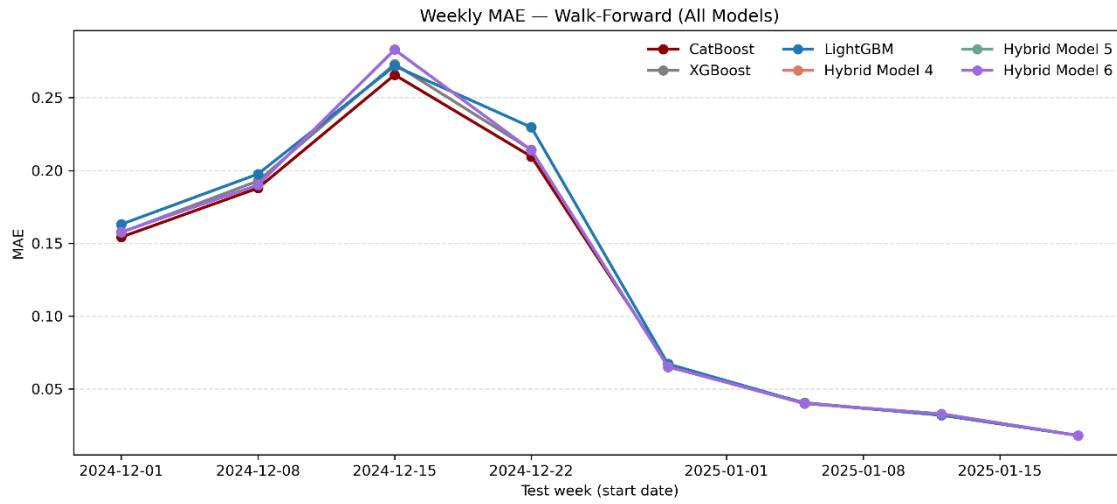


Figure 10: Walk-Forward MAE for the five forecasting models (Dec 2024 – Jan 2025)

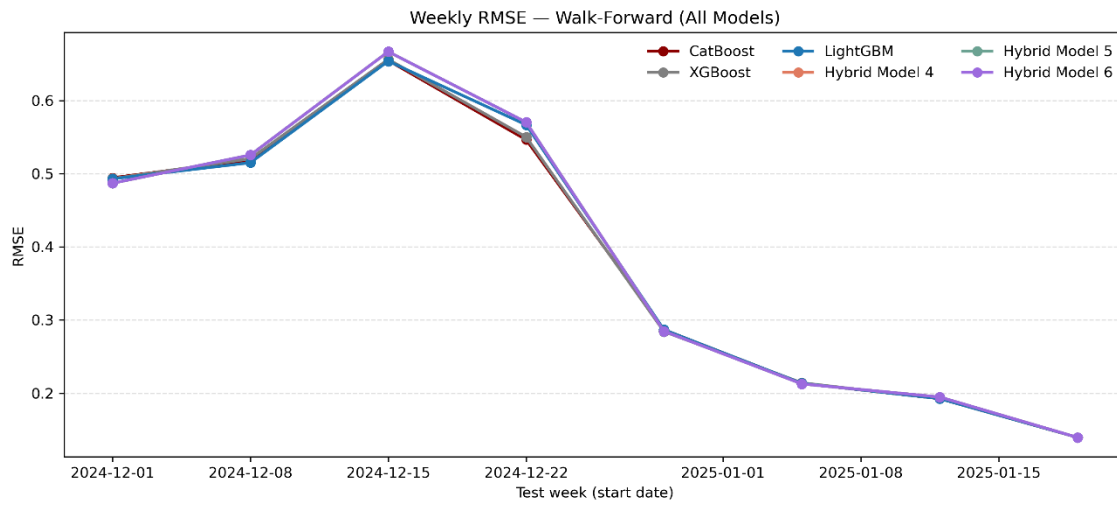


Figure 11: Walk-Forward RMSE for the five forecasting models (Dec 2024 – Jan 2025)

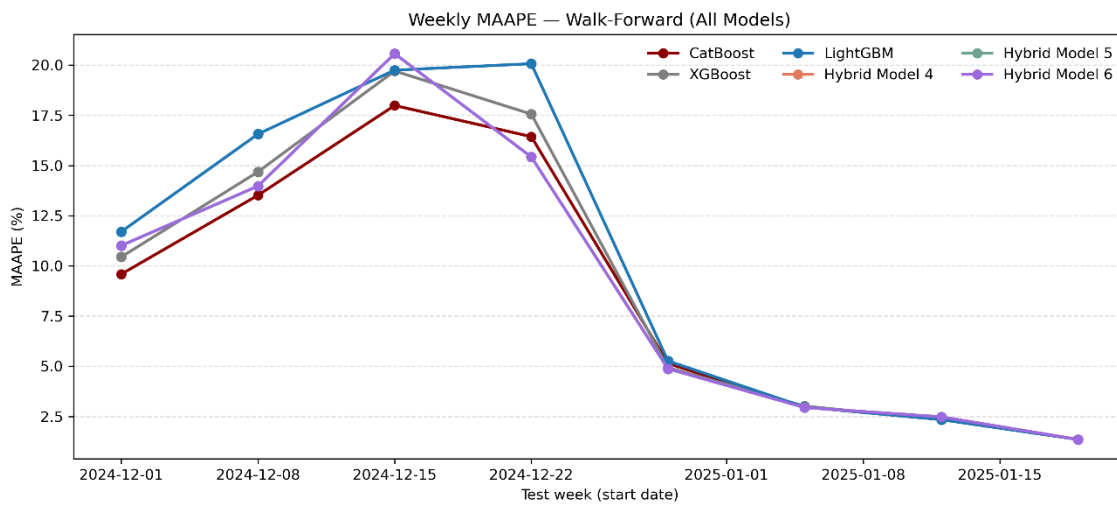


Figure 12: Walk-Forward MAAPE for the five forecasting models (Dec 2024 – Jan 2025)

In the context of weekly walk-forward cross-validation, the relative ranking of models changes compared to the hold-out evaluation. CatBoost emerges as the best-performing algorithm across all three metrics, achieving an MAE of 0.1218, an RMSE of 0.3809 and an MAAPE of 8.67%. It is followed by XGBoost, with slightly higher error values (0.1241, 0.3814 and 9.25%). The three hybrid configurations (Models 4, 5 and 6) yield identical results, with an MAE of 0.1251, an RMSE of 0.3852 and a MAAPE of 9.08%, positioning them just behind XGBoost but ahead of LightGBM. LightGBM records the weakest performance, with the highest errors among all models (0.1274, 0.3827 and 10.0%). It is important to note that error values increased in the weeks leading up to Christmas, reflecting the challenges of capturing demand that surges related to holiday-driven purchasing behaviour. Moreover, it is precisely in these high-error weeks that the differences between the models become more evident, whereas in January, when errors are lower, the performance of the various methods appears nearly indistinguishable.

This shift in ranking is particularly noteworthy when contrasted with the hold-out results, where LightGBM demonstrated strong performance, even surpassing XGBoost. The decline in LightGBM's performance under cross-validation indicates that, while it adapts well to a specific fixed test window, it struggles with consistency when assessed across rolling horizons. In contrast, the hybrid models, although less accurate overall than CatBoost and XGBoost, exhibited greater stability than LightGBM. This suggests that their fallback mechanisms, such as moving averages or NB components, effectively help to mitigate instability in low-demand or highly intermittent series.

In conclusion, the cross-validation results indicate that CatBoost and XGBoost are the most dependable models concerning both accuracy and stability, with CatBoost exhibiting the strongest generalisation across various weeks. The hybrid models displayed moderate robustness but did not outperform the standalone boosting algorithms. Meanwhile, LightGBM demonstrated a weaker capacity for generalisation under rolling evaluation. Collectively, these findings underscore the significance of supplementing hold-out tests with cross-validation to assess not only the accuracy but also the stability of models in practical operational settings.

4.2. Comparison with the Company's Current Forecasting Model

To benchmark against the company's existing forecasting methodology, CatBoost and XGBoost were selected as the reference models. CatBoost consistently emerged as the top performer across both evaluation strategies, demonstrating its superior accuracy and robustness. However, the second-best model varied depending on the evaluation method: LightGBM outperformed XGBoost in the hold-out test, but its performance declined significantly under cross-validation, where XGBoost demonstrated greater stability and secured the second position. Although the hybrid models were competitive, they did not outperform the boosting algorithms. Therefore, CatBoost and XGBoost were deemed the most suitable models for comparison with the company's current forecasting approach.

As previously described, the company's current strategy relies on a short-term replenishment method that extrapolates sales velocity from the most recent 7 to 14 days, incorporating adjustments for seasonal variations. While this approach offers a straightforward and operationally efficient solution, it is predominantly reactive and heavily reliant on very recent sales trends.

To ensure a fair comparison, it was crucial to align the outputs of the machine learning models with the company's forecasting framework. While CatBoost and XGBoost provide daily forecasts, the company generates its forecasts based on a scheduled date and a predefined rotation cycle. Based on this information, coverage intervals were established by determining a start and an end date that define the period for which the company's forecast remains valid. As a result, the daily predictions from the machine learning models were aggregated to correspond with these specified intervals. For example, consider a scenario where a company releases a forecast on January 1, accompanied by a rotation cycle of 7 days. In this case, the company's forecast reflects the sales for the period spanning January 2 to January 8. So, the machine learning forecasts generated for each of those seven days were aggregated, yielding a total forecast value that is directly comparable to the company's figure for the same time frame.

Along with forecast values, the actual sales were accumulated over the corresponding coverage periods. This allowed for direct comparisons between model predictions, company forecasts and real outcomes on the same temporal basis, using the test period from December 2024 to January 2025.

For evaluation purposes, the error metrics consistently utilised throughout this study, namely, MAE, RMSE and MAAPE, were applied. In addition to these aggregate measures, a distributional analysis of forecast errors was performed. This analysis included histograms that displayed the frequency of under and overpredictions, as well as summary tables detailing the number of instances in which forecasts underestimated, overestimated or precisely matched the actual sales.

Additionally, a new metric called the Weighted Score was introduced. This measure differentiates between underprediction and overprediction, aligning with the company's operational priorities. Specifically, underpredictions, which indicate lost sales, were given twice the weight of overpredictions, which relate to excess stock. This adjustment was implemented to better reflect the practical implications of forecasting errors within the context of retail replenishment.

The comparative analysis commenced with an examination of the overall error metrics. As illustrated in Table 15 (the highest-performing value for each measure is emphasised in bold), the company's forecasts exhibited significantly higher errors across all measures, whereas both boosting models demonstrated markedly improved accuracy. CatBoost consistently outperformed the others, with XGBoost closely following, confirming the superiority of boosting techniques over the company's reactive short-term forecasting method.

Table 15: Global error metrics comparison between company forecasts and boosting models

Model	MAE	RMSE	MAAPE
Company forecast	2.800	4.234	143.875%
CatBoost	0.100	0.635	2.905%
XGBoost	0.118	0.752	3.192%

Following the presentation of the aggregate error metrics, the analysis was expanded to examine the distribution of signed forecast errors. Figure 13 depicts the frequency of errors associated with the company's forecasting method alongside the two boosting models. The results reveal notable differences in their behaviour: the company's forecasts demonstrate a wide dispersion of errors, including significant deviations from actual sales, while both CatBoost and XGBoost exhibit distributions that are highly concentrated around zero. This concentration underscores the superior accuracy and stability of the machine learning models, which not only yield a significantly higher number of exact matches but

also experience fewer severe deviations.

For completeness, Appendix A reports the full distribution of signed errors across all models. This table reinforces the graphical evidence, showing that CatBoost and XGBoost consistently generate the majority of exact predictions, while the company's method is associated with a higher incidence of overestimations.

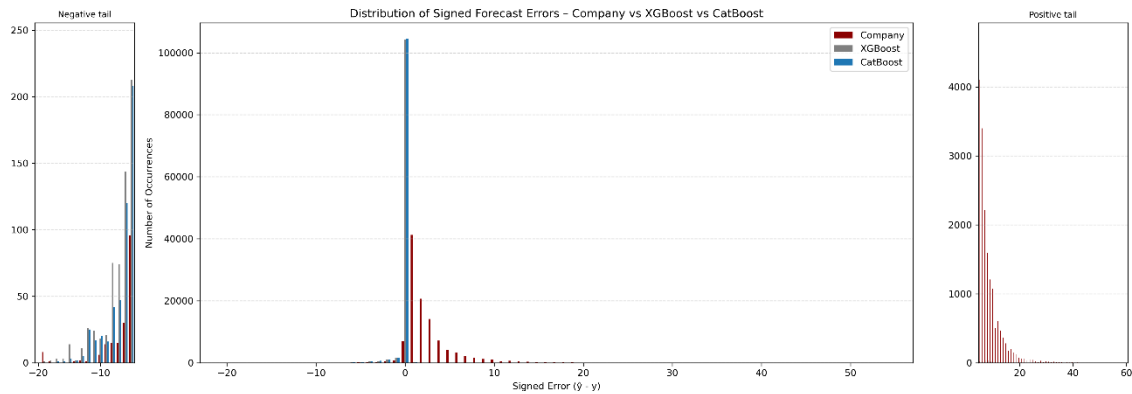


Figure 13: Comparative distribution of signed forecast errors across the company's current forecasting method, XGBoost and CatBoost

To complement the analysis of error magnitudes, Table 16 (best outcome for each case is highlighted in bold) presents the distribution of forecasts across three categories, underpredictions, overpredictions and exact matches, along with the corresponding Weighted Score. This metric penalises underpredictions twice as heavily as overpredictions, reflecting the company's operational reality where stockouts are more detrimental than excess inventory.

Table 16: Distribution of forecast outcomes (counts and percentages) and Weighted Scores

Model	Underprediction	Exact match	Overprediction	Weighted Score
Company forecast	1716 (1.58%)	7083 (6.50%)	100133 (91.92%)	103565
CatBoost	4198 (3.85%)	104438 (95.87%)	296 (0.27%)	8692
XGBoost	4195 (3.85%)	104273 (95.72%)	464 (0.43%)	8854

The results reveal that the company's forecasting method is highly skewed towards overprediction, with more than 91% of intervals exceeding actual sales and only 6.5% achieving an exact match. In contrast, both CatBoost and XGBoost exhibit a markedly different behaviour, with more than 95% of their forecasts corresponding exactly to realised sales. Although both models display a small proportion of underpredictions ($\approx 3.8\%$), their

incidence of overprediction is negligible (below 0.5%).

The Weighted Score provides important insights into the error structure. The company's forecasts are often characterised by overpredictions, which are usually less costly than underpredictions; however, the high frequency of these errors leads to a significant penalty. In contrast, both CatBoost and XGBoost demonstrate markedly lower scores. Although there is a greater emphasis on underpredictions in these models, the overall number of errors they generate is considerably reduced, resulting in costs that remain well below those associated with the company's forecasts. This finding reinforces the conclusion that the boosting models not only minimise the average magnitude of errors but also lessen the practical implications of forecasting mistakes.

Figure 14 presents a visual summary of the results, showcasing the relative distribution of underpredictions, overpredictions and exact matches across the models. This graphical representation reinforces the findings detailed in **Erro! A origem da referência não foi encontrada.**, emphasising the notable tendency of the company's forecasts toward overestimation and the significant occurrence of exact matches observed in both CatBoost and XGBoost.

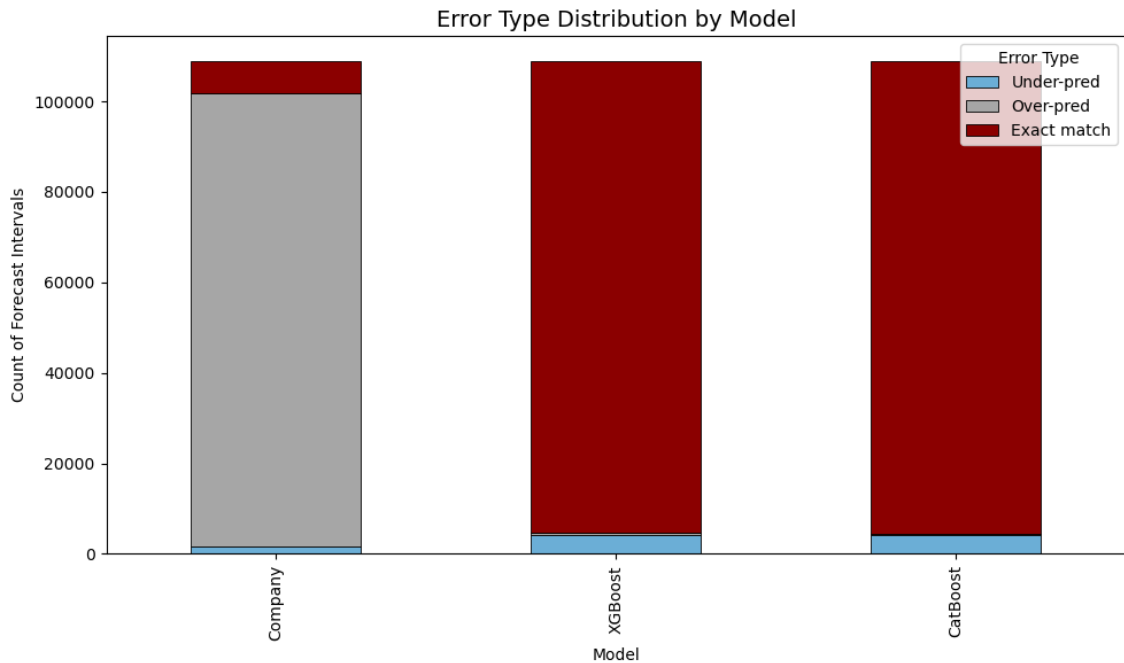


Figure 14: Error type distribution (underpredictions, overpredictions and exact matches) for company forecasts, XGBoost and CatBoost

The comparative evaluation reveals that the company's replenishment-based forecasting method, despite its operational simplicity, consistently results in higher error

magnitudes and exhibits a significant bias towards overestimation. In contrast, both CatBoost and XGBoost achieve notably lower error rates and provide a more balanced error profile, characterised by a greater number of exact matches and significantly reduced Weighted Scores. These findings underscore the superiority of boosting models over the company's current methodology, offering not only enhanced predictive accuracy but also improved alignment with practical decision-making needs in retail replenishment.

Beyond the fixed 2 - 1 weighting scheme used operationally, a sensitivity analysis was conducted by generalising the formulation of the metric. Instead of fixing the penalty for underprediction at $X = 2$ and for overprediction at 1, the weighted score was expressed as:

$$WS(X) = X \times U + 1 \times O$$

where U and O denote the total magnitude of under and overpredictions, respectively, and X is the adjustable penalty factor for underpredictions.

The objective was to determine the value of X for which the company's forecasts would achieve the same weighted score as the boosting models. This threshold corresponds to the solution of:

$$X^* \times U_{company} + O_{company} = X^* \times U_{CatBoost\ or\ LightGBM} + O_{CatBoost\ or\ LightGBM}$$

which can be rearranged as:

$$X^* = \frac{O_{CatBoost\ or\ LightGBM} - O_{company}}{U_{company} - U_{CatBoost\ or\ LightGBM}}$$

Using the exact error magnitudes reported in Appendix A, the thresholds were found to be $X^* = 39.74$ for the comparison with XGBoost and $X^* = 45.64$ for the comparison with CatBoost. These results imply that, as long as the penalty assigned to underpredictions remains below these values (i.e., $X < 40$ or $X < 46$), both boosting models outperform the company's method. The company's forecasts would only become preferable in terms of weighted score if penalties for underpredictions were set unrealistically high.

5. Conclusions

This chapter summarises the main findings of the study and discusses their practical and academic implications. It starts by highlighting the key achievements of the research, then addresses its limitations and concludes by suggesting potential directions for future work.

5.1. Achievements

Sales forecasting plays a crucial role in informed decision-making within the fast fashion industry, characterised by short product life cycles, unpredictable demand patterns and swiftly changing consumer preferences. The primary objective of this dissertation was to design and assess forecasting models aimed at improving the replenishment process for a Portuguese fashion retail company. The goal was to reduce the risks of stockouts and overstocks while ensuring that the proposed solution could be seamlessly integrated into the company's operational framework, thereby facilitating more efficient and data-driven inventory management.

The empirical analysis demonstrated that using advanced machine learning models significantly enhances forecasting accuracy compared to the company's current replenishment method, which primarily relies on extrapolating recent sales trends. In particular, CatBoost consistently showed the best performance in both hold-out testing and walk-forward validation, while XGBoost also proved to be a stable and competitive option. By incorporating a broader range of explanatory variables, including temporal, product-related and store-level features, these models effectively captured demand dynamics beyond just short-term fluctuations.

The research successfully developed a forecasting framework specifically designed to address the operational characteristics of fast fashion retail. The models demonstrated their effectiveness in tackling challenges such as intermittent demand, a high frequency of zero-sales observations and short product life cycles. A stratified evaluation based on product life-cycle length indicated that forecast accuracy improves with the availability of longer sales histories, while still exceeding baseline performance in the more demanding short-lifecycle segment.

Another significant achievement pertains to operational feasibility. The results

indicated that the baseline configurations of the boosting algorithms were adequate for producing accurate and robust forecasts, thereby eliminating the need for extensive hyperparameter optimisation. This equilibrium between predictive accuracy and computational efficiency guarantees that the models are suitable for integration into retail operations, where timely forecasts are essential for informed decision-making.

A key achievement of this study is its potential for scalability. Although the empirical analysis focused on a single product category and five stores, the modelling framework is designed to accommodate future expansion. However, extending the methodology to additional categories and retail locations will require careful adjustments, particularly for products with longer life cycles or more diverse demand patterns. Nonetheless, the results derived from this study lay a strong foundation for such adaptations, illustrating that advanced machine learning models can effectively enhance replenishment and inventory practices at scale.

Considering the results across accuracy, operational feasibility and the potential for adaptation, the study provides evidence that advanced forecasting methods can enhance the company's replenishment process and support more efficient inventory management in fast fashion retail.

5.2. Limitations

Like any empirical study, this dissertation has several limitations that should be considered when interpreting the results. Some limitations stem from the methodological choices made to ensure the feasibility of the research, while others are related to broader constraints concerning data, modelling approaches and computational resources.

The research was intentionally focused on a specific product category and five selected stores. This decision was made pragmatically to ensure that the dataset could be effectively prepared and processed within the constraints of available time and resources. Similarly, the chosen temporal window was established with the same intent: to provide a sufficient coverage of sales history while maintaining the feasibility of implementing multiple forecasting models. While these choices facilitated the successful execution of the project, they also indicate that the findings cannot be directly generalised to encompass the entire product portfolio or retail network of the company.

The dataset posed several challenges that served as limitations. Firstly, the number

of products with long life cycles was restricted, which constrained the analysis of forecasting performance for this particular segment. Additionally, the sales data were sparse and exhibited a highly intermittent nature, characterised by frequent occurrences of zero-demand observations. While this data structure is representative of the fast fashion context, it limited the effectiveness of certain traditional methods and added complexity to the modelling process.

Methodological limitations were also noted in the study. The focus was solely on the product–store level of analysis, without exploring alternative granularities such as aggregated store or category-level forecasts, which could have provided additional insights. Although hybrid strategies were implemented, they did not yield the expected improvements, highlighting the challenges of effectively combining statistical and machine learning approaches in the context of irregular demand. Moreover, hyperparameter tuning resulted in only modest gains, partly due to the sparse and heterogeneous nature of the dataset. Additionally, the use of a fixed 18-month horizon for all models, while aligning with the project's objectives, may not adequately capture the flexibility needed in more dynamic operational environments.

Computational constraints posed a significant limitation to the study. The implementation of a rolling daily forecasting approach, which would allow for predictions to be updated from the very first day of sales, was not feasible. Additionally, the study's scope was confined to traditional statistical methods, machine learning algorithms and hybrid approaches. Advanced deep learning architectures, which are increasingly being utilised in forecasting literature, were not incorporated into the analysis. Future research could benefit from exploring these models, especially in terms of managing complex temporal dependencies.

Overall, while these limitations do not diminish the validity of the results, they do emphasise the boundaries of the study. The findings are based on a specific product category within a single company; thus, further research is necessary before generalising them to other contexts within the fast fashion retail sector.

5.3. Future Work

Several promising avenues for future research could build upon the findings of this dissertation. A key direction is to expand the dataset beyond the single product category and

the five stores utilised in this study. Such an expansion would enable the testing of the proposed methodology's scalability across a broader and more diverse range of conditions. Furthermore, it would allow for an assessment of whether the modelling approach maintains its effectiveness when applied to larger datasets. In this context, specific adjustments will be necessary during the modelling phase. For instance, the division between training and testing sets may need to be modified for products with longer life cycles. This is just one example of the adjustments required, as expanding the dataset would likely necessitate further refinements throughout various stages of the forecasting process.

Another promising avenue for development is the implementation of a dynamic "day-by-day" forecasting strategy. In contrast to the fixed-horizon approach utilised in this study, this framework would continuously update predictions as new sales data becomes available. For instance, the model could be retrained after the first day of sales to project the second day, then use the data from the first two days to forecast the third and so forth. This rolling methodology would enable the company to achieve increasingly accurate forecasts in real time, thereby minimising the need to wait for extended observation periods before generating predictions. Such an approach would be particularly advantageous for products with short life cycles, where timely and precise forecasting is essential for effective stock management.

Future research may also investigate more sophisticated modelling techniques. In particular, deep learning methods have gained considerable attention in the forecasting field and may offer the ability to identify more complex temporal patterns compared to traditional or tree-based machine learning methods. With enhanced computational resources, these approaches could be assessed to determine whether they provide advantages in contexts characterised by intermittent and irregular demand.

Incorporating additional external variables presents a valuable opportunity to enhance the accuracy of forecasting. Research has shown that factors like weather conditions, macroeconomic indicators and consumer sentiment can significantly influence retail demand (e.g., Choi et al., 2014; Fildes et al., 2022). While this dissertation has already taken into account some external variables, such as marketing campaigns and promotional activities, future research could delve deeper into the benefits of including these additional variables. This could lead to a more nuanced understanding of demand fluctuations and improve the alignment of forecasts with the fundamental drivers of the market.

Collectively, these strategies present a research agenda to enhance the reliability of

forecasting models and ensure their better alignment with the operational realities of the fast fashion retail sector.

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Appendix

Appendix A – Full Distribution of Forecast Errors

Error	Company	XGBoost	CatBoost
-19	8	1	0
-18	1	2	0
-17	0	3	1
-16	0	3	1
-15	0	14	3
-14	1	2	2
-13	2	11	5
-12	1	26	25
-11	0	24	17
-10	6	18	20
-9	14	21	16
-8	15	75	42
-7	15	74	47
-6	30	144	120
-5	96	213	208
-4	122	386	376
-3	183	578	611
-2	441	1026	1022
-1	781	1574	1682
0	7083	104273	104438
1	41263	256	215
2	20630	39	54
3	14134	15	22
4	7138	27	4
5	4109	25	1
6	3400	21	0
7	2211	30	0
8	1590	33	0
9	1206	15	0
10	1078	3	0
11	499	0	0
12	603	0	0
13	469	0	0
14	360	0	0
15	283	0	0
16	172	0	0

17	205	0	0
18	144	0	0
19	120	0	0
20	77	0	0
21	64	0	0
22	60	0	0
23	36	0	0
24	44	0	0
25	40	0	0
26	26	0	0
27	10	0	0
28	27	0	0
29	11	0	0
30	17	0	0
31	15	0	0
32	13	0	0
33	14	0	0
34	10	0	0
35	11	0	0
36	7	0	0
37	2	0	0
38	5	0	0
39	7	0	0
40	10	0	0
41	3	0	0
42	6	0	0
46	1	0	0
52	1	0	0
53	1	0	0
58	1	0	0

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